



TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
PULCHOWK CAMPUS

*An Adaptive Overcurrent Protection Scheme for Dual-Setting Directional
Recloser and Fuse Coordination in Unbalanced Distribution Networks
With Distributed Generation*

BY:

Jhala Nath Kafle

(081MSPSE009)

A PROJECT REPORT SUBMITTED IN PARTIAL FULFILLMENT OF THE
REQUIREMENTS FOR THE MASTER'S DEGREE IN POWER SYSTEM
ENGINEERING

DEPARTMENT OF ELECTRICAL ENGINEERING

OCTOBER, 2026

PAGE OF APPROVAL

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
PULCHOWK CAMPUS
DEPARTMENT OF ELECTRICAL ENGINEERING

The undersigned certify that they have read, and recommended to the Institute of Engineering for acceptance, a project report entitled “*An Adaptive Overcurrent Protection Scheme for Dual-Setting Directional Recloser and Fuse Coordination in Unbalanced Distribution Networks With Distributed Generation*” submitted by *Jhala Nath Kafle* in partial fulfilment of the requirements for the Master’s degree in Power System Engineering.

Jeetendra Chaudhary
Head of Department &
M.Sc. Program Coordinator (MSPDE)
Department of Electrical Engineering

Akhileshwar Mishra
M.Sc. Program Coordinator (MSPSE)
Department of Electrical Engineering

DATE OF APPROVAL: 6 October 2026

COPYRIGHT

The author has agreed that the Library, Department of Electrical Engineering, Pulchowk Campus, Institute of Engineering may make this report freely available for inspection. Moreover, the author has agreed that permission for extensive copying of this project report for scholarly purpose may be granted by the Head of the Department wherein the project report was done. It is understood that recognition will be given to the author of this report and to the Department of Electrical Engineering, Pulchowk Campus, Institute of Engineering in any use of the material of this project report. Copying or publication or the other use of this report for financial gain without approval of the Department of Electrical Engineering, Pulchowk Campus, Institute of Engineering and the author's written permission is prohibited.

Request for permission to copy or to make any other use of the material in this report in whole or in part should be addressed to:

Head
Department of Electrical Engineering
Pulchowk Campus, Institute of Engineering
Lalitpur, Nepal

ACKNOWLEDGEMENTS

I express my sincere gratitude to the Head of the Department and to the faculty of the Department of Electrical Engineering, Pulchowk Campus, Institute of Engineering, for the guidance, the review comments and the facilities that made this project possible, and to the coordinator of the M.Sc. programme in Power System Engineering for the encouragement throughout the work.

I also thank my colleagues of the M.Sc. Power System Engineering programme for the discussions on distribution protection and on the use of DIgSILENT PowerFactory, and my family for their patience and support.

Jhala Nath Kafle
081MSPSE009

ABSTRACT

The recloser–fuse coordination of distribution networks with distributed generation (DG) can be lost when conventional overcurrent protection is used, because the DG changes the magnitude and the direction of the fault current seen by the protective devices. This project studies a protection coordination strategy based on a dual-setting directional recloser (DSDR) for the fuse-saving scheme of an unbalanced distribution network. The mid-line recloser is given independent forward and reverse settings, so that it coordinates with the lateral fuses for faults on both sides of it. The strategy is implemented on the IEEE 13-node test feeder in DIgSILENT PowerFactory with a synchronous DG, the characteristics of commercial recloser relays and fuses, unbalanced load-flow and short-circuit studies, and an electromagnetic-transient simulation of the reclosing sequence. The results show that the DG makes the fuses operate before the fast operation of a conventional recloser, increasingly so as the DG penetration rises. With the DG connected, a conventional single-setting mid-line recloser keeps the coordination in 24 of the 39 studied fault cells. The dual-setting directional recloser alone raises this to 25 cells, and together with a revision of the fuse sizes to 38 of 39 cells under the zero-margin criterion; the remaining cell, an LG fault at node 692, is limited by the delayed dial range of the recloser. When a three-cycle breaker interrupting time and a fuse safety margin are included, the number of coordinated cells falls to 31 and 26. The relay and fuse models of PowerFactory confirm the calculated operating times within 1.22 per cent.

Keywords: distributed generation, dual-setting directional recloser, fuse saving, recloser–fuse coordination, IEEE 13-node feeder, DIgSILENT PowerFactory.

TABLE OF CONTENTS

PAGE OF APPROVAL	II
COPYRIGHT	III
ACKNOWLEDGEMENTS	IV
ABSTRACT	V
TABLE OF CONTENTS	VI
LIST OF FIGURES	IX
LIST OF TABLES	X
LIST OF SYMBOLS AND ABBREVIATIONS	XII
CHAPTER ONE: INTRODUCTION	1
1.1 Background	1
1.2 Problem Statement	1
1.3 Objectives	1
1.3.1 General Objective	1
1.3.2 Specific Objectives	2
1.4 Scope of the Study	2
1.5 Significance of the Study	2
1.6 Limitations of the Study	2
1.7 Organization of the Report	3
CHAPTER TWO: LITERATURE REVIEW	4
2.1 Recloser–Fuse Protection of Distribution Feeders	4
2.2 Impact of Distributed Generation and Directional Adaptive Protection	4
2.3 Relay and Fuse Characteristics	4
2.4 Related Research and Data Used	4
2.5 Method Followed and Research Gap	5
CHAPTER THREE: METHODOLOGY	6
3.1 Research Methodology	6
3.2 Model Setup in DIGSILENT PowerFactory	7
3.3 Load Flow Analysis	7
3.4 Recloser Pickup Current	7
3.5 Short-Circuit Analysis	7
3.6 Mathematical Formulation of the DSDR	7
3.6.1 Forward Fast and Delayed Operating Time	7
3.6.2 Forward Pickup Current and Constraints	8
3.6.3 Fuse Characteristic and Coefficient	8

3.6.4	Series Fuses	8
3.6.5	Reverse Operating Times, Pickup and Constraints	9
3.7	Device Characteristics Used in the Model	9
3.7.1	R1: GE IAC77B801A	9
3.7.2	R2: GE/Alstom CDG34	9
3.7.3	Fuses	10
3.8	Coordination Criteria	10
3.9	Design Procedure	10
CHAPTER FOUR: SYSTEM MODELING AND SIMULATION SETUP		12
4.1	DIgSILENT PowerFactory Environment	12
4.2	IEEE 13-Node Feeder Model	12
4.3	Source and Feeder	13
4.4	Distributed Generation Model	13
4.5	Recloser Model	14
4.6	Fuse Model	14
4.7	Fault Model	15
4.8	Measurement and Data Acquisition	16
4.9	Simulation Cases	16
CHAPTER FIVE: RESULTS OBTAINED		17
5.1	Load-Flow Results	17
5.2	Branch Currents and Fault Levels	18
5.3	Recloser Pickup Currents and Settings	18
5.4	Short-Circuit Results With and Without DG	18
5.5	Fuse Coefficients	19
5.6	Conventional Coordination Without DG	20
5.7	Coordination With DG and a Conventional R2	21
5.8	DSDR Settings and Fuse Revision	23
5.9	Coordination With the DSDR	24
5.10	Operating Times With the DSDR	26
5.11	Single Setting Against Dual Setting on Selected Faults	27
5.12	Time-Sequence Check	29
5.13	Time-Domain Verification	30
5.14	Verification With the Relay and Fuse Models of PowerFactory	32
5.15	Sensitivity of the Result to the Margin	32
5.16	Effect of the DG Penetration Level	32
5.17	Results Summary	36
CHAPTER SIX: DISCUSSION		37
6.1	Effect of DG on Fault Current	37
6.2	Effect of Fault Current Direction	37
6.3	Performance of the Conventional Protection	37
6.4	Performance of the DSDR	37

6.5	Limits Reached by the Devices	37
6.6	Engineering Implication	38
CHAPTER SEVEN: CONCLUSION AND FUTURE WORK		39
7.1	Conclusion	39
7.2	Major Findings	39
7.3	Contributions of the Present Work	40
7.4	Future Work	40
REFERENCES		41
APPENDIX A: COMPLETE COPY OF THE REFERENCE PAPER		42
APPENDIX B: IMPLEMENTATION FORMULAS		43
APPENDIX C: COMPARISON WITH THE REFERENCE PAPER		44

LIST OF FIGURES

Figure 3.1	Flowchart of the recloser–fuse coordination method with the DSDR.	6
Figure 4.1	IEEE 13-node feeder with the reclosers R1 and R2, the fuses and the DG at node 692.	12
Figure 5.1	LG fault at 611: R2 forward, fuses F684 and F671-2; CTI = +143 ms.	21
Figure 5.2	LLG fault at the middle of 692–675 through 1 Ω ; CTI = +127 ms. .	21
Figure 5.3	Coordination status with DG, without the DSDR.	22
Figure 5.4	LL fault at 646 through 1 Ω : R1 with F646 and F632; CTI = +200 ms.	22
Figure 5.5	LL fault at 645 through 1.5 Ω , conventional R2.	23
Figure 5.6	Three-phase fault at 10 per cent of line 632–633, conventional R2; CTI = –83 ms, the fuse melts first.	23
Figure 5.7	The two setting groups of R2: reverse (DG contribution, faults towards 632) and forward (grid contribution, faults beyond 671), each with one fault of the model.	24
Figure 5.8	Coordination status with DG and the DSDR.	25
Figure 5.9	Bolted LL fault at 646: R2 reverse setting with F646 and F632; CTI = +328 ms.	25
Figure 5.10	Bolted LL fault at 646: R1 with F646 and F632; CTI = +226 ms. . .	26
Figure 5.11	Three-phase fault at 10 per cent of line 632–633: single setting (left, CTI = –22 ms) and dual setting (right, CTI = +16 ms).	28
Figure 5.12	LG fault at 646: the single-setting R2 does not pick up the reverse current (left); the reverse group does (right, CTI = +661 ms).	29
Figure 5.13	Time-sequence check of all 39 cells, single against dual setting. . .	30
Figure 5.14	Time-domain current at node 632 for an LL (a–c) fault at 684 through 0.2 Ω : without DG (top) and with DG (bottom).	31
Figure 5.15	Percentage variation of the CTI with DG penetration (outer ring 671, inner ring 633).	33
Figure 5.16	Currents through the fuse (grid and DG share), the recloser (grid share) and from the DG, and the resulting CTI, against DG penetration.	34
Figure 5.17	Time–current characteristics at all penetration levels: fuse operating points (squares) and recloser operating points (circles).	34
Figure 5.18	Node 633, fuse F633 and R1, at each penetration level, with the short-circuit currents, operating times and CTI.	35
Figure C.1	CTI against DG penetration: this study (left) and the values read from the paper’s Fig. 7 (right).	51

LIST OF TABLES

Table 4.1	Parameters of the synchronous DG in the PowerFactory model. . . .	13
Table 4.2	Recloser settings. For R2 the fast plug is half and the delayed plug equal to $1.25I_{nom}$, because the CDG curve starts at twice the plug setting.	14
Table 4.3	Fuses and their sizes (A055C, E-rated). Bold: changed by the fuse revision with the DG.	15
Table 4.4	Simulation cases.	16
Table 5.1	Bus voltages from the unbalanced load flow, without and with the DG.	17
Table 5.2	Load current at the reclosers (largest phase; + is forward, away from the grid).	17
Table 5.3	Rated, minimum and maximum fault current of the protected branches without DG. XFM1 HV side: 4.16 kV currents for a fault on the 0.48 kV side.	18
Table 5.4	Maximum fault current and recloser currents without and with the DG (+ forward, – reverse).	19
Table 5.5	Fuse coefficients, eq. (3.7), $a_i = -1.8$. Size and t_{MMT} : fuse of the design without DG at the same current.	20
Table 5.6	Coordination status with DG and a conventional R2. ✓ exists, × does not exist, – not applicable.	22
Table 5.7	Coordination status with DG and the revised fuses: single-setting R2 against dual-setting R2.	24
Table 5.8	Recloser operating times with the DSDR, DG in service.	26
Table 5.9	Fuse times with the DSDR, DG in service. Margin = t_{MMT} minus the last fast trip (R2 for faults below R2; the later of R1 and R2 reverse above it). Nodes 632, 671 and 680 are on the main feeder and have no fuse.	27
Table 5.10	Nine faults with the DG in service: single-setting against dual-setting R2, same fuses.	27
Table 5.11	Three-phase fault at 10 per cent of line 632–633: currents, settings and operating times.	28
Table 5.12	LG fault at 646 (phase c): currents, settings and operating times. . .	29
Table 5.13	Reclosing sequence of the EMT simulation.	31
Table 5.14	CTI against DG penetration, conventional protection.	33
Table 5.15	Largest bolted fault current at each node (A) against DG penetration.	35
Table 5.16	Load flow, pickups by eq. (3.3) and lowest fault current through each recloser against DG penetration. *: below the start of the recloser's curve with the fixed settings (not detected).	36
Table 5.17	Summary of the coordination results.	36
Table C.1	Comparison of the results with the reference paper.	44
Table C.2	DG parameters: reference paper (Table I) against the PowerFactory model.	46

Table C.3	Branch currents: model against Table II of the paper. XFM1 HV side: the model values are 4.16 kV currents for a fault on the 0.48 kV side, so this row is not comparable.	47
Table C.4	Maximum fault current, DG out: IEEE benchmark, this study and the paper's Table II. Deviations are relative to the benchmark.	48
Table C.5	Fuse coefficients against Table III of the paper. " b_i rev.": index i counted from the source, which reproduces the paper's values in R1's zone.	48
Table C.6	Recloser operating times against Table IV of the paper.	49
Table C.7	Fuse times against Table IV of the paper.	49
Table C.8	Coordination status with DG and a conventional R2: model against the paper's Fig. 14.	50
Table C.9	Reclosing sequence of the EMT simulation against the paper's Fig. 10.	50
Table C.10	CTI shares against DG penetration: this study and the paper's Fig. 7.	51

LIST OF SYMBOLS AND ABBREVIATIONS

Symbols

I_f	Fault current seen by a protection device
I_p	Pickup current of a recloser
I_s	Plug (current) setting of the CDG relay
I_{nom}	Rated (load) current
M	Multiple of pickup, I_f/I_p
TDS, TMS	Time-dial / time-multiplier setting
t_F, t_D	Fast and delayed operating time of a recloser
t_{MMT}, t_{TCT}	Minimum-melting and total-clearing time of a fuse
A, B, n	Constants of the inverse-time curve
a_i, b_i	Coefficients of the fuse characteristic
k, l, m	Index of the recloser, of a downstream and of an upstream fault location
z, i	Number of series fuses in the fault path; position of a fuse in it
Z_f	Fault impedance
X_d, X'_d, X''_d	Direct-axis synchronous, transient and subtransient reactance
X_q, X'_q, X''_q	Quadrature-axis reactances
R_a	Armature resistance

Abbreviations

CT	Current Transformer
CTI	Coordination Time Interval
DG	Distributed Generation
DSDR	Dual-Setting Directional Recloser
EMT	Electromagnetic Transient (simulation)
LG	Line-to-Ground fault
LL	Line-to-Line fault
LLG	Line-to-Line-to-Ground fault
LLL	Three-Phase fault
MMT	Minimum Melting Time
OLF	Overload Factor
PD	Protection Device
PF	DIgSILENT PowerFactory
R1	Feeder-Head Recloser
R2	Mid-Line Recloser (the DSDR)
RMS	Root Mean Square
SBDG	Synchronous-Based Distributed Generator
TCC	Time–Current Characteristic
TCT	Total Clearing Time
TDS	Time Dial Setting
TMS	Time Multiplier Setting

CHAPTER ONE: INTRODUCTION

1.1 Background

Overcurrent protection of radial distribution networks is traditionally coordinated on the assumption that the fault current flows from the substation towards the loads. A recloser upstream of a fuse first operates on a fast characteristic, to clear a temporary fault before the fuse melts, and then on a delayed characteristic, to let the fuse clear a permanent fault. This *fuse-saving* arrangement only works while the fuse curve lies between the two recloser curves over the whole range of fault currents.

Distributed generation changes this. A DG downstream of a recloser feeds current towards a fault upstream of it, so the recloser carries current in the reverse direction, and a fuse between the two sources carries the sum of both contributions while each recloser carries only one. The current seen by a protection device then depends on the fault location, the topology and the DG, and a single fixed setting is no longer suitable for both directions.

The dual-setting directional recloser (DSDR) addresses this by giving the mid-line recloser separate pickup and time-dial settings for the forward and for the reverse direction. This project implements the DSDR method of [1] on the IEEE 13-node feeder in DIgSILENT PowerFactory and evaluates it.

1.2 Problem Statement

The problem addressed is the loss of recloser–fuse coordination caused by the change of fault-current magnitude and direction when a DG is connected. A conventional scheme that is coordinated without DG may lose the fuse-saving property in the zone between the grid and the DG. The project investigates whether the DSDR method restores the coordination in an independently built model that uses the manufacturer-based time–current characteristics of the specified commercial protection devices.

1.3 Objectives

1.3.1 General Objective

To build and validate a DIgSILENT PowerFactory implementation of the DSDR protection-coordination method on the IEEE 13-node test feeder, with and without distributed generation.

1.3.2 Specific Objectives

1. Build and verify the IEEE 13-node feeder with its recloser and fuse protection.
2. Perform unbalanced load-flow and short-circuit studies for all fault types and phase combinations.
3. Calculate the recloser pickup currents, time dials and the fuse coefficients with the method's equations.
4. Establish the conventional recloser–fuse coordination without DG.
5. Connect the 4.05 MVA synchronous DG at node 692 and quantify the change of the fault currents and of their direction at the reclosers.
6. Classify the coordination with DG for a conventional recloser R2.
7. Design the forward and reverse settings of R2 as a DSDR, classify the resulting coordination, and determine the corresponding operating times.
8. Verify the reclosing sequence in the time domain.
9. Study how the coordination degrades as the DG penetration rises from 0 to 100 per cent.

1.4 Scope of the Study

The study covers the IEEE 13-node feeder: unbalanced load flow, LG, LL, LLG and LLL faults at twelve fault locations, pickup and time-dial calculation, fuse coordination, the DSDR forward and reverse settings, the coordination classification with and without the DSDR, and one EMT simulation. The IEEE 34-node feeder is outside the completed work.

1.5 Significance of the Study

The study gives a reproducible implementation of a protection-coordination method for feeders with DG in a tool that utilities use. It shows what the method achieves when the real relay dials and fuse sizes are respected, and it separates the contribution of the dual setting from the contribution of the fuse revision.

1.6 Limitations of the Study

- The method does not uniquely specify the R2 settings, the fuse sizes, the source impedance or the fault type of each operating-time case; these choices were therefore

defined in this study, and every choice and its basis is listed in Section 4.8.

- The coordination is classified with a zero margin; the effect of a breaker interrupting time and a fuse safety margin is evaluated separately in Section 5.15.
- The library model of the CDG34 relay is not directional. The two setting groups are modelled as separate relay units and the direction is assigned from the fault location.
- Coordination is judged with a zero-margin criterion (the fuse must not start to melt before the fast trip). Section 5.15 shows how the result changes with a breaker time and a fuse margin.
- The DG penetration study (Section 5.16) uses one DG location (692) and keeps the protection of the design without DG at every level.
- The IEEE 34-node feeder has not been modelled.

1.7 Organization of the Report

Chapter One introduces the problem and the objectives. Chapter Two reviews the literature on the protection concepts and the method followed. Chapter Three gives the method and its equations. Chapter Four describes the PowerFactory model, the DG, the protection devices and the simulation cases. Chapter Five presents the results. Chapter Six discusses the findings and their limits. Chapter Seven gives the conclusions and the future work. The comparison with the reference paper is in Appendix C.

CHAPTER TWO: LITERATURE REVIEW

2.1 Recloser–Fuse Protection of Distribution Feeders

Distribution feeders are protected by reclosers and fuses with inverse-time characteristics: the operating time falls as the multiple of the pickup current rises, so the device nearest the fault acts first and the upstream devices act as backup [2]. The pickup fixes where the characteristic starts and the time dial scales the operating time; real relays offer only discrete taps and a limited dial range. In the fuse-saving scheme the recloser’s fast curve must lie below the minimum-melting curve of the fuse, so that a temporary fault is cleared without blowing the fuse, and its delayed curve above the total-clearing curve, so that the fuse isolates a permanent fault. Fuses in series must be selective: the fuse nearer the fault clears before the upstream fuse starts to melt.

2.2 Impact of Distributed Generation and Directional Adaptive Protection

A DG raises the current through the fuses and can reverse the current through a recloser. The recloser and the fuse then no longer carry the same current, and a scheme that was coordinated without DG may let the fuse melt before the fast trip [3]. A directional element distinguishes the direction of the fault current, and adaptive protection changes the settings with the state of the network. The dual-setting directional recloser (DSDR) combines both: it holds a forward and a reverse setting group and selects one from the direction of the current, without communication [1].

2.3 Relay and Fuse Characteristics

Inverse-time relay curves are given by standard equations [2] or by the manufacturer’s own equation or table; the GE IAC and the CDG series follow their manufacturer’s curves, in which the time is the time dial multiplied by a function of $M = I_f/I_p$ [4]. A fuse is described by its minimum-melting time (MMT), when the element starts to melt, and its total-clearing time (TCT), when the arc is extinguished.

2.4 Related Research and Data Used

A classification technique decides for each fault whether recloser–fuse coordination with DG exists [3]. A benchmark overcurrent protection scheme for the IEEE 13-node feeder [5], with device types and fuse sizes, is given in [6]; the published short-circuit cur-

rents of the IEEE test feeders [7] are used here to validate the fault levels, and the relay curves are those of the manufacturers' manuals [4]. Because the feeder is unbalanced, all calculations are made per phase and keep only the phases that exist at each node.

2.5 Method Followed and Research Gap

The study follows the adaptive overcurrent protection scheme with a DSDR of [1]: its equations for the forward and reverse operating times, pickups, fuse coefficients and setting constraints, and its flowchart for the design sequence. The reference reports that the DSDR restores coordination in every case, but it does not publish the DSDR settings, and its fuse curves differ from its own fuse equation. The gap addressed here is an independent implementation in PowerFactory with device-specific relay curves, dial ranges and fuse sizes, which shows how much of the improvement comes from the dual setting and where the method reaches the limits of the devices.

CHAPTER THREE: METHODOLOGY

3.1 Research Methodology

The work follows the flowchart of the reference method [1], shown in Figure 3.1 as it is implemented. It has three stages: (A) the conventional scheme without DG, (B) the same settings with the DG connected, and (C) R2 as a dual-setting directional recloser with the fuse revision of the method.

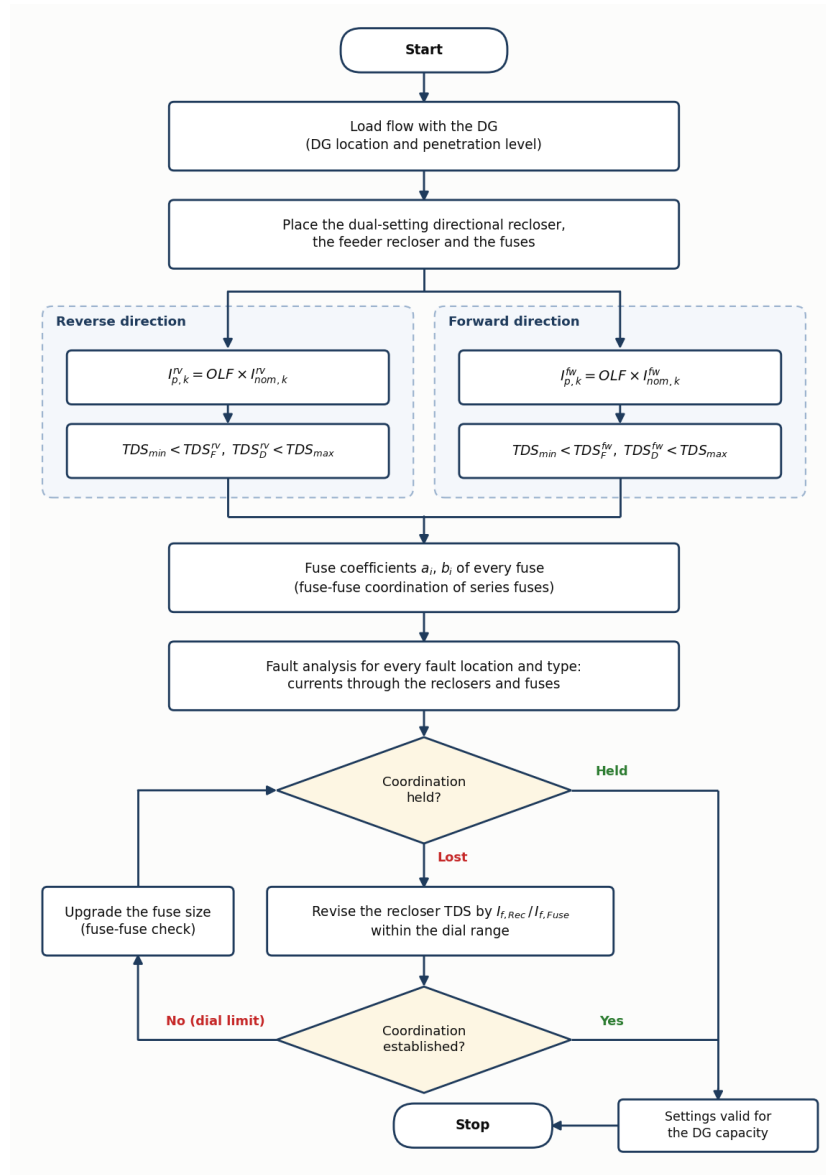


Figure 3.1 Flowchart of the recloser-fuse coordination method with the DSDR.

3.2 Model Setup in DIgSILENT PowerFactory

The study is carried out in DIgSILENT PowerFactory 2021 SP2 [8]. The feeder is the IEEE 13-node model of the DIgSILENT example library; its lines, loads and regulator taps were checked against the IEEE data. The study case that includes the 115/4.16 kV, 5 MVA substation transformer is used, because it reproduces the IEEE short-circuit benchmark [7].

3.3 Load Flow Analysis

The load flow is the unbalanced three-phase AC load flow of PowerFactory. It gives the rated current of every protected branch (largest phase), the bus voltages per phase, and the current and its direction at the two reclosers with and without the DG.

3.4 Recloser Pickup Current

The method uses an overload factor of 1.25 [1]. The pickup is 1.25 times the rated current of the protected branch, eq. (3.3), and is then set on the nearest available tap of the relay.

3.5 Short-Circuit Analysis

Short circuits are calculated with the *complete method* of PowerFactory, which starts from the pre-fault load flow. LG, LL, LLG and LLL faults are applied at every node for every phase combination that exists there, with the DG out and in. The maximum fault current of a branch is a bolted fault at its nearest node; the minimum is an LG fault through $3\ \Omega$ at its farthest node [1]. The current through R1, R2, every fuse and every branch of Table 5.3 is stored for each fault.

3.6 Mathematical Formulation of the DSDR

The equations below are those of the reference method [1].

3.6.1 Forward Fast and Delayed Operating Time

$$t_{k,l,F}^{fw} = TDS_{k,F}^{fw} \left[\frac{A_k^{fw}}{\left(I_{f,l}/I_{p,k}^{fw} \right)^{n_k^{fw}} - 1} + B_k^{fw} \right] \quad (3.1)$$

$$t_{k,l,D}^{fw} = TDS_{k,D}^{fw} \left[\frac{A_k^{fw}}{\left(I_{f,l} / I_{p,k}^{fw} \right)^{n_k^{fw}} - 1} + B_k^{fw} \right] \quad (3.2)$$

The fast and the delayed operation of recloser k for a fault at location l use the same curve with two time dials. $I_{f,l}$ is the fault current through the recloser and I_p its pickup.

3.6.2 Forward Pickup Current and Constraints

$$I_{p,k}^{fw} = OLF \times I_{nom,k}^{fw} \quad (3.3)$$

$$TDS_{k,min} < TDS_{k,F}^{fw}, TDS_{k,D}^{fw} < TDS_{k,max} \quad (3.4)$$

$$I_{p,k,min} < I_{p,k}^{fw} < I_{p,k,max} \quad (3.5)$$

The pickup lies above the load current and below the minimum fault current of the zone; the dials must stay inside the range of the device.

3.6.3 Fuse Characteristic and Coefficient

$$\log t_{fuse,i} = a_i \log I_{f,i} + b_i \quad (3.6)$$

$$b_i = \log \left[\frac{t_{k,F}^{fw} + t_{k,D}^{fw}}{2} \right] - a_i \log I_{f,i} \quad (3.7)$$

with $a_i = -1.8$. Equation (3.7) places a single fuse midway between the fast and the delayed time at the calibration current.

3.6.4 Series Fuses

$$t_{TCT,i} \leq 0.75 t_{MMT,i+1} \quad (3.8)$$

$$b_i = \log \left[t_{k,F}^{fw} + \frac{i \left(t_{k,D}^{fw} - t_{k,F}^{fw} \right)}{z + 1} \right] - a_i \log I_{f,i} \quad (3.9)$$

where z is the number of series fuses in the fault path and $i = 1$ is the fuse closest to the fault. Equation (3.9) divides the window between the two recloser times among the series fuses.

3.6.5 Reverse Operating Times, Pickup and Constraints

$$t_{k,m,F}^{rv} = TDS_{k,F}^{rv} \left[\frac{A_k^{rv}}{\left(I_{f,m}/I_{p,k}^{rv} \right)^{n_k^{rv}} - 1} + B_k^{rv} \right] \quad (3.10)$$

$$t_{k,m,D}^{rv} = TDS_{k,D}^{rv} \left[\frac{A_k^{rv}}{\left(I_{f,m}/I_{p,k}^{rv} \right)^{n_k^{rv}} - 1} + B_k^{rv} \right] \quad (3.11)$$

$$I_{p,k}^{rv} = OLF \times I_{nom,k}^{rv} \quad (3.12)$$

$$TDS_{k,min} < TDS_{k,F}^{rv}, TDS_{k,D}^{rv} < TDS_{k,max} \quad (3.13)$$

$$I_{p,k,min} < I_{p,k}^{rv} < I_{p,k,max} \quad (3.14)$$

Here m is a fault location upstream of the recloser. The reverse current is the contribution of the DG, and the reverse pickup is based on the reverse load current that flows when the DG exports towards the substation.

3.7 Device Characteristics Used in the Model

The model uses the time–current curves of the PowerFactory library types of the protection devices.

3.7.1 R1: GE IAC77B801A

The operating time of R1 follows the GE IAC extremely inverse characteristic. The manufacturer gives it as $t = M_{TDS} [A + B/(M - C) + D/(M - C)^2 + E/(M - C)^3]$ with $A = 0.004$, $B = 0.638$, $C = 0.620$, $D = 1.787$ and $E = 0.246$ [4]. With the constants of the PowerFactory library type (the same values to four digits) the curve is

$$t = TDS \left[0.004 + \frac{0.6379}{M - 0.62} + \frac{1.7872}{(M - 0.62)^2} + \frac{0.2461}{(M - 0.62)^3} \right], \quad M = \frac{I_f}{I_p}, \quad (3.15)$$

valid from $M = 1.5$ to $M = 40$. For 4219 A, for example, it gives 0.096 s (fast, TDS 0.5) and 1.926 s (delayed, TDS 10).

3.7.2 R2: GE/Alstom CDG34

The CDG extremely inverse characteristic is a manufacturer's table of time against $M = I_f/I_s$ for time multipliers from 0.1 to 1.0. It starts at twice the plug setting and is

interpolated on log-log axes:

$$t = t_1 \left(\frac{t_2}{t_1} \right)^f, \quad f = \frac{\ln(M/M_1)}{\ln(M_2/M_1)}. \quad (3.16)$$

The table is not defined below a multiplier of 0.1, so 0.1 is the fastest setting of this relay. The dial of R1 is called the time dial setting (TDS), as in the IAC77B801A documentation, whereas the CDG34 documentation calls the dial of R2 the time multiplier setting (TMS). Both scale the operating time of the curve in the same way, and the general equations of Section 3.6 write TDS for either.

3.7.3 Fuses

The fuses are Gould-Shawmut A055C (5.5 kV, E-rated) with their minimum-melting and total-clearing curves. The fuse times are read from these curves at the current through the fuse with the same interpolation as eq. (3.16). The straight line of eq. (3.6) is used only to compute the coefficients of Table 5.5.

3.8 Coordination Criteria

A fault cell is classified as *coordination exists* when all of the following hold:

$$t_{MMT}(I_{fuse}) > t_F \quad \text{for every recloser that feeds the fault through the fuse,} \quad (3.17)$$

$$t_{TCT}(I_{fuse}) < t_D \quad \text{for the same reclosers,} \quad (3.18)$$

together with the series-fuse condition of eq. (3.8) and a margin of at least 10 cycles between the delayed curves of R2 and R1. The coordination time interval is $CTI = t_{MMT} - t_F$, the time by which the fuse starts to melt after the latest fast trip; a cell is held when $CTI > 0$ (the zero-margin criterion). An additional engineering margin, for the breaker interrupting time and for uncertainty, is not part of the classification and is examined separately in Section 5.15. A cell (node and fault type) is held only if every phase combination of that fault type is held. The fuse and each recloser are evaluated at their own currents, because with DG they differ.

3.9 Design Procedure

The flowchart of Figure 3.1 is applied in these steps.

1. Run the load flow without DG and set the recloser pickups from the rated currents.
2. Run the short-circuit study for the minimum and maximum fault currents of each sec-

tion.

3. Set the dials: R1 at TDS 0.5 and 10; R2 fast at its lowest dial, delayed 10 cycles below R1.
4. Calculate the fuse coefficients and check the installed fuses, including the series fuses.
5. Classify the coordination without DG and revise the fuses until every case is held.
6. Connect the DG with the settings unchanged and classify again.
7. Set the reverse pickup and the reverse dials of R2 as a DSDR.
8. Revise the dials and, where a dial limit is reached, the fuse sizes until the cases are held.
9. Classify the coordination with the DSDR and determine the operating times.
10. Verify with PowerFactory's own relay and fuse models and the EMT simulation.

4.3 Source and Feeder

The source is an external grid at 115 kV behind the 5 MVA, 115/4.16 kV substation transformer ($u_k = 8.06\text{percent}$). With this source the fault currents agree with the IEEE benchmark [7] within about 2 per cent at every node. The alternative study case with a stiff 4.16 kV source overstates the fault levels by 20 to 65 per cent and is not used. The regulator taps are fixed at +10, +8 and +11.

4.4 Distributed Generation Model

The DG is a 4.05 MVA, 0.69 kV synchronous machine connected to node 692 through a 0.69/4.16 kV transformer with a leakage reactance of 0.15 p.u. It is modelled with the standard synchronous-machine model of PowerFactory, round rotor, with the short-circuit data of Table 4.1 (values read back from the model). In the load flow the machine delivers 3.24 MW and controls its terminal voltage to 1.0 p.u.

Parameter	Symbol	Value	Unit
Leakage reactance	X_l	0.05	pu
Stator resistance	R_a	0.0014	pu
d-axis synchronous reactance	X_d	1.4	pu
d-axis transient reactance	X'_d	0.231	pu
d-axis subtransient reactance	X''_d	0.118	pu
d-axis transient open-circuit time constant	T'_{d0}	5.5	s
d-axis subtransient open-circuit time constant	T''_{d0}	0.05	s
q-axis synchronous reactance	X_q	1.372	pu
q-axis transient reactance	X'_q	0.8	pu
q-axis subtransient reactance	X''_q	0.118	pu
q-axis transient open-circuit time constant	T'_{q0}	1.25	s
q-axis subtransient open-circuit time constant	T''_{q0}	0.19	s
Mechanical starting time	$M = 2H$	1.5	s
DG rating	S_n	4.05	MVA
DG voltage	U_n	0.69	kV
Step-up transformer, HV side	U_{HV}	4.16	kV
Step-up transformer leakage reactance	x_T	0.15	pu

Table 4.1 Parameters of the synchronous DG in the PowerFactory model.

4.5 Recloser Model

R1 is at the feeder head (RG60–632). R2 is on line 632–671 at the 671 end; it is the DSDR, because with the DG at 692 it carries forward current for faults at 671 and beyond and reverse current for faults towards 632. Table 4.2 gives the settings. The forward and the reverse group of R2 are four relay units of the CDG34 library type on two current transformers.

Recloser	Relay, CT	$1.25 I_{nom}$	Setting	Dial fast / delayed
R1 forward	GE IAC77B801A, 900/5	734.6 A	tap 4 A = 720 A	TDS 0.5 / 10
R2 forward	CDG34, 1000/5	587.5 A	plug 300 A / 600 A	TMS 0.1 / 1.0
R2 reverse	CDG34, 500/5	315.3 A	plug 150 A / 300 A	TMS 0.1 / 1.0

Table 4.2 Recloser settings. For R2 the fast plug is half and the delayed plug equal to $1.25 I_{nom}$, because the CDG curve starts at twice the plug setting.

4.6 Fuse Model

Table 4.3 lists the fifteen fuses. The starting size is the size drawn in the figures of the reference paper where a figure shows the fuse (the fuse bands were digitised and matched to the A055C library curves within 2 per cent), otherwise the size of the benchmark scheme [6]. The last two columns are the sizes after the fuse revision of the method without DG and with the DSDR.

Fuse	Location	Start	No DG	DSDR	Source of the starting size
F632	lateral 632–645, at 632	400E	400E	500E	Ref. [1], Figs. 12, 15, 16
F633	lateral 632–633, at 632	250E	250E	400E	Benchmark scheme [6], Table V
F634	XFM-1, 0.48 kV side	gL-800A	gL-800A	gL-800A	Benchmark scheme [6], Table V
F645	load at 645	250E	250E	250E	Benchmark scheme [6], Table V
F646	line 645–646, at 645	300E	300E	400E	Ref. [1], Figs. 11, 15, 16
F-DL	distributed-load lateral on 632–671	250E	250E	400E	Benchmark scheme [6], Table V
F671	load at 671	250E	250E	250E	Benchmark scheme [6], Table V
F671-1	671–692 (switch), at 671	300E	300E	400E	Ref. [1], Fig. 9
F692	load at 692	200E	200E	200E	Benchmark scheme [6], Table V
F692-R	cable 692–675, at 692	250E	200E	250E	Ref. [1], Fig. 9
F675	load at 675	200E	200E	200E	Benchmark scheme [6], Table V
F671-2	lateral 671–684, at 671	300E	300E	300E	Ref. [1], Fig. 8
F684	line 684–611, at 684	200E	200E	200E	Ref. [1], Fig. 8
F611	load at 611	125E	125E	125E	Benchmark scheme [6], Table V
F652	cable 684–652, at 684	150E	150E	150E	Benchmark scheme [6], Table V

Table 4.3 Fuses and their sizes (A055C, E-rated). Bold: changed by the fuse revision with the DG.

4.7 Fault Model

Faults are applied at the nodes 632, 633, 634, 645, 646, DL, 671, 692, 675, 680, 684, 611 and 652. Only the fault types that the phases of a node allow are applied, for example LL and LLG on phases b–c at 645 and 646, and LG only at 611 and 652. This gives 39 node/fault-type cells. Selected faults through a fault impedance are added for the time–current figures.

4.8 Measurement and Data Acquisition

For every fault the phase currents through R1, R2 and each fuse are read from the short-circuit result. The operating current of a device is its largest phase current. The forward direction is away from the grid; R2 sits at the 671 end of its line, where PowerFactory counts the current positive into the line, so its values are multiplied by -1 . The following modelling choices were made:

- the source: the IEEE benchmark substation transformer, with the complete short-circuit method;
- R1: tap 4 A on a 900/5 current transformer, i.e. a pickup of 720 A;
- R2: all settings of Table 4.2, derived from eqs. (3.3) and (3.12);
- the operating-time cases of Table 5.8: bolted LLL faults, LL at two-phase nodes, LG at single-phase nodes;
- the EMT study: a reclosing dead time of 0.2 s and a breaker time of three cycles.

4.9 Simulation Cases

Case	DG	Faults	Purpose
Load flow	out / in	–	Rated currents, voltages, direction at R1 and R2
Maximum faults	out / in	All nodes, LG/LL/LLG/LLL, all phase combinations, bolted	Classification, Tables II–IV
Minimum faults	out / in	LG through $3\ \Omega$ at all nodes	$I_{f,min}$, pickup check
Figure faults	in	Selected faults, bolted and through a fault impedance	Time–current figures
EMT	out / in	LL a–c at 684 through $0.2\ \Omega$	Reclosing sequence
Penetration	0–100 %	LLL at 633, LL at 684, all nodes	CTI, fault levels and pickups against DG size

Table 4.4 Simulation cases.

CHAPTER FIVE: RESULTS OBTAINED

5.1 Load-Flow Results

Table 5.1 gives the bus voltages with and without the DG. Without DG the lowest voltage is 0.971 p.u. (phase c at 611). The DG raises the voltages of the loaded phases a and c at node 671 and beyond by about 0.03 to 0.05 p.u.

Bus	Without DG (p.u.)			With DG (p.u.)		
	A	B	C	A	B	C
650	0.9981	1.0170	0.9979	1.0084	1.0200	1.0057
RG60	1.0605	1.0679	1.0665	1.0715	1.0710	1.0748
632	1.0187	1.0616	1.0142	1.0482	1.0640	1.0361
633	1.0158	1.0588	1.0124	1.0454	1.0611	1.0344
634	0.9918	1.0406	0.9936	1.0221	1.0430	1.0161
645	–	1.0540	1.0106	–	1.0563	1.0325
646	–	1.0522	1.0085	–	1.0546	1.0304
DL	1.0028	1.0666	0.9939	1.0407	1.0685	1.0221
671	0.9877	1.0722	0.9753	1.0334	1.0738	1.0094
692	0.9877	1.0722	0.9753	1.0334	1.0738	1.0094
675	0.9809	1.0767	0.9729	1.0271	1.0781	1.0072
680	0.9877	1.0722	0.9753	1.0334	1.0738	1.0094
684	0.9859	–	0.9732	1.0315	–	1.0074
652	0.9811	–	–	1.0265	–	–
611	–	–	0.9712	–	–	1.0055

Table 5.1 Bus voltages from the unbalanced load flow, without and with the DG.

Table 5.2 shows the load current at the two reclosers. With the DG the current through R2 reverses: 252.2 A flows from 671 towards 632. This reverse load current is the basis of the reverse pickup, eq. (3.12).

Device	Without DG (A)	Direction	With DG (A)	Direction
R1	+587.7	forward	+284.9	forward
R2	+470.0	forward	–252.2	reverse

Table 5.2 Load current at the reclosers (largest phase; + is forward, away from the grid).

5.2 Branch Currents and Fault Levels

Table 5.3 gives the rated current of every protected branch from the unbalanced load flow, the minimum fault current (LG through $3\ \Omega$ at the farthest node of the branch's section) and the maximum fault current (bolted, nearest node). The largest fault current at the feeder head is 4.73 kA; the fault levels agree with the IEEE 13-node short-circuit benchmark within about 2 per cent.

Branch	I_{nom} (A)	$I_{f,min}$ (kA)	$I_{f,max}$ (kA)
RG60–632	587.7	1.07	4.73
632–633	81.0	0.76	4.1
632–645	143.1	0.73	3.42
632–671	470.0	0.88	3.39
XFM1-HV side	81.0	0.08	1.81
XFM1-LV side	706.4	0.71	15.67
645–646	66.1	0.73	3.11
671–692	229.7	0.74	3.39
671–684	71.1	0.66	2.66
671–680	0.0	0.64	2.93
692–675	205.9	0.74	3.15
684–611	71.1	0.68	1.86
684–652	63.0	0.66	1.87

Table 5.3 Rated, minimum and maximum fault current of the protected branches without DG. XFM1 HV side: 4.16 kV currents for a fault on the 0.48 kV side.

5.3 Recloser Pickup Currents and Settings

With the rated currents of Table 5.3, eq. (3.3) gives 734.6 A for R1 and 587.5 A for R2 forward. The settings actually applied are those of Table 4.2. Without DG both pickups are below the lowest minimum fault current of their zones (1074 A and 879 A).

5.4 Short-Circuit Results With and Without DG

Table 5.4 gives the largest fault current at each node and the current through the two reclosers. The DG raises the fault current at every node, by 38 per cent at 632 and by 64 per cent at 671. For faults at 632, 633, 645, 646 and DL the current through R2 becomes negative: R2 then carries only the DG contribution, 1.1 to 2.0 kA, in the reverse direction, while R1 still carries the grid contribution.

Bus	Fault	$I_{f,max}$ at the fault (A)		Through R1 (A)		Through R2 (A)	
		no DG	DG	no DG	DG	no DG	DG
632	LLL ABC	4735	6526	+4735	+4734	+0	−1813
633	LLL ABC	4102	5426	+4162	+3970	+66	−1500
634	LLL ABC	15671	17608	+2034	+1611	+298	−618
645	LLG BC	3418	4341	+3528	+3330	+478	−1262
646	LLG BC	3112	3854	+3192	+3000	+475	−1135
671	LLL ABC	3390	5564	+3413	+3412	+3390	+3391
692	LLL ABC	3390	5564	+3413	+3412	+3390	+3391
675	LLL ABC	3151	5022	+3202	+3090	+3174	+3050
680	LLL ABC	2930	4460	+3006	+2789	+2976	+2757
684	LLG AC	2655	4107	+2836	+2669	+2744	+2651
652	LG A	1874	2239	+2094	+1795	+2059	+1761
611	LG C	1857	2236	+2084	+1814	+1991	+1718
DL	LLL ABC	3946	5902	+3959	+3954	+1	−1972

Table 5.4 Maximum fault current and recloser currents without and with the DG (+ forward, − reverse).

5.5 Fuse Coefficients

Table 5.5 gives the fuse coefficients of eq. (3.9), with $i = 1$ for the fuse closest to the fault. The last columns give the installed fuse and its melting time at the same current; it is much shorter than the time the equation asks for, because real fuse curves are steeper than the slope -1.8 .

Fuse	I_f (A)	i/z	t_F (s)	t_D (s)	t_{fuse} (s)	b_i	Size	t_{MMT} (s)
F632	3418	2/2	0.133	2.670	1.824	6.62	400E	0.512
F633	4102	1/1	0.100	2.008	1.054	6.53	250E	0.102
F634	15671	1/2	0.439	8.772	3.217	8.06	gL-800A	0.021
F645	3418	1/2	0.133	2.670	0.979	6.35	250E	0.158
F646	3112	1/2	0.156	3.115	1.142	6.35	300E	0.273
F-DL	3946	1/1	0.106	2.130	1.118	6.52	250E	0.112
F671	3390	1/1	0.054	1.601	0.827	6.27	250E	0.161
F671-1	3390	3/3	0.054	1.601	1.214	6.44	300E	0.218
F692	3390	1/2	0.054	1.601	0.569	6.11	200E	0.094
F692-R	3151	2/3	0.057	1.829	0.943	6.27	200E	0.112
F675	3151	1/3	0.057	1.829	0.500	6.00	200E	0.112
F671-2	2655	3/3	0.069	2.572	1.946	6.45	300E	0.428
F684	1857	2/3	0.113	5.778	2.946	6.35	200E	0.465
F611	1857	1/3	0.113	5.778	1.530	6.07	125E	0.120
F652	1874	1/2	0.112	5.660	1.961	6.18	150E	0.207

Table 5.5 Fuse coefficients, eq. (3.7), $a_i = -1.8$. Size and t_{MMT} : fuse of the design without DG at the same current.

5.6 Conventional Coordination Without DG

With the starting fuse sizes 35 of the 39 cells are coordinated. The four lost cells are the faults at 675, where the series fuses F692-R (250E) and F671-1 (300E) are not selective: F692-R does not clear the fault early enough before F671-1 starts to melt. The fuse revision of the method changes F692-R to 200E, after which all 39 cells are coordinated. Figures 5.1 and 5.2 show two cases: R2's fast curve lies below the fuses and its delayed curve above them. In every time–current plot of this chapter the dotted lines carry each operating point to the time axis with its operating time. The arrow at the fuse current gives the coordination time interval $CTI = t_{MMT} - t_F$ (green when positive, red when the fuse melts first) and the margin $t_D - t_{TCT}$ by which the fuse clears before the delayed trip.

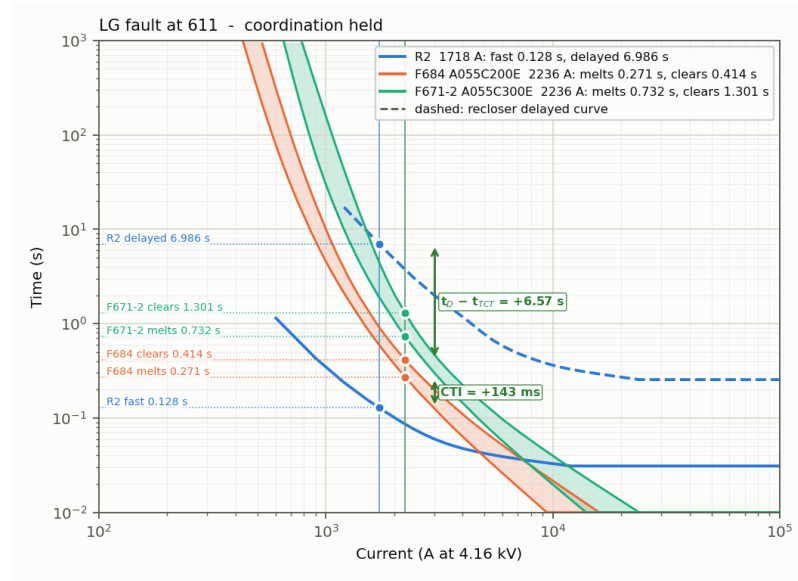


Figure 5.1 LG fault at 611: R2 forward, fuses F684 and F671-2; CTI = +143 ms.

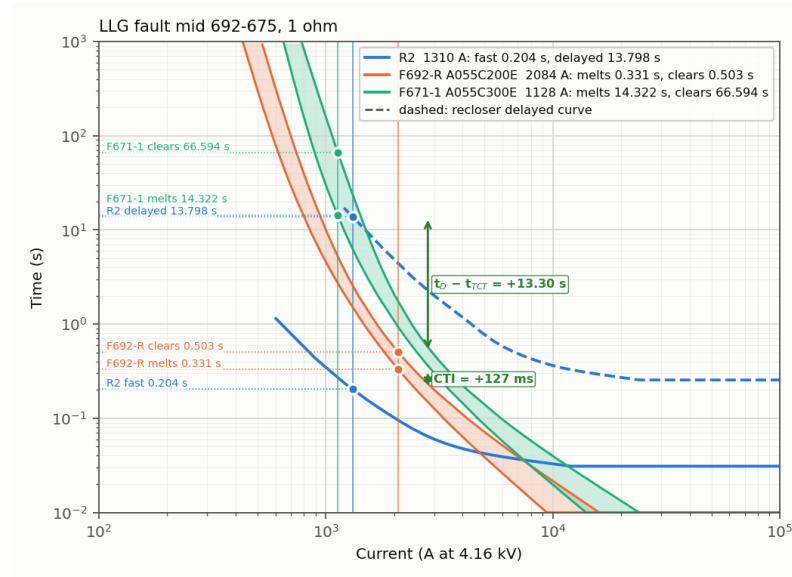


Figure 5.2 LLG fault at the middle of 692–675 through 1 Ω ; CTI = +127 ms.

5.7 Coordination With DG and a Conventional R2

With the DG connected and the settings unchanged, coordination is held in 24 of 39 cells (Table 5.6, Figure 5.3). The lost cells are at 633, 645, 646, DL and 675: the fuse now carries the grid and the DG current and melts before the fast trip, and for some LG faults the single-setting R2 does not pick up the reverse DG current at all. At 633 and DL the fuses have almost no margin even without DG, so every fault type is lost there.

Fault	632	633	645	646	DL	671	692	675	680	684	611	652
LG	✓	×	×	×	×	✓	✓	✓	✓	✓	✓	✓
LL	✓	×	✓	×	×	✓	✓	×	✓	✓	—	—
LLG	✓	×	✓	×	×	✓	✓	×	✓	✓	—	—
LLL	✓	×	—	—	×	✓	✓	×	✓	—	—	—

Table 5.6 Coordination status with DG and a conventional R2. ✓ exists, × does not exist, — not applicable.

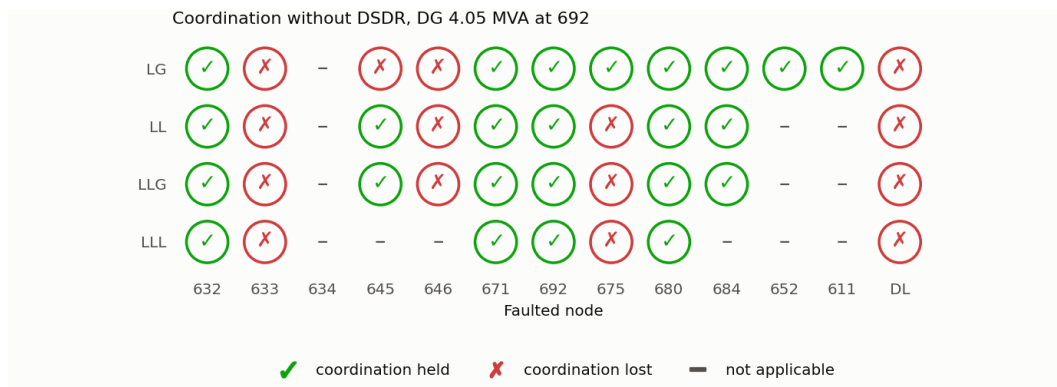


Figure 5.3 Coordination status with DG, without the DSDR.

Figures 5.4 to 5.6 show three cases with the conventional R2.

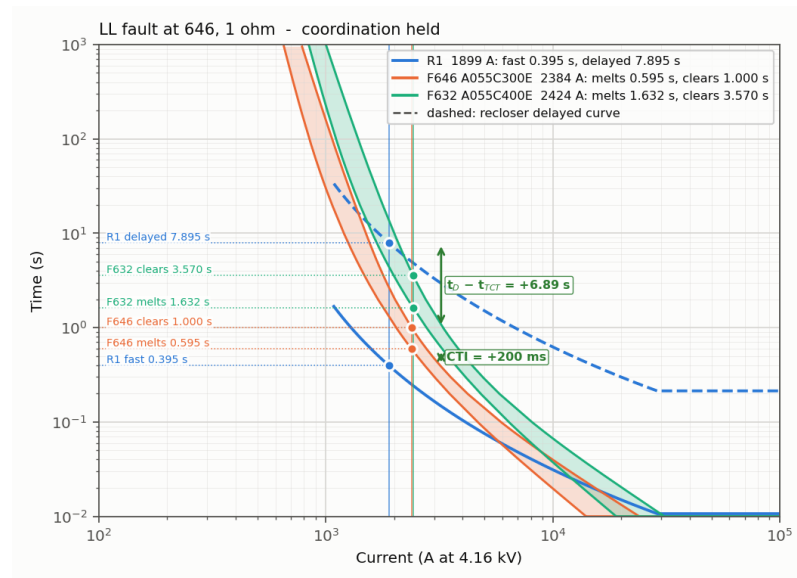


Figure 5.4 LL fault at 646 through 1 Ω: R1 with F646 and F632; CTI = +200 ms.

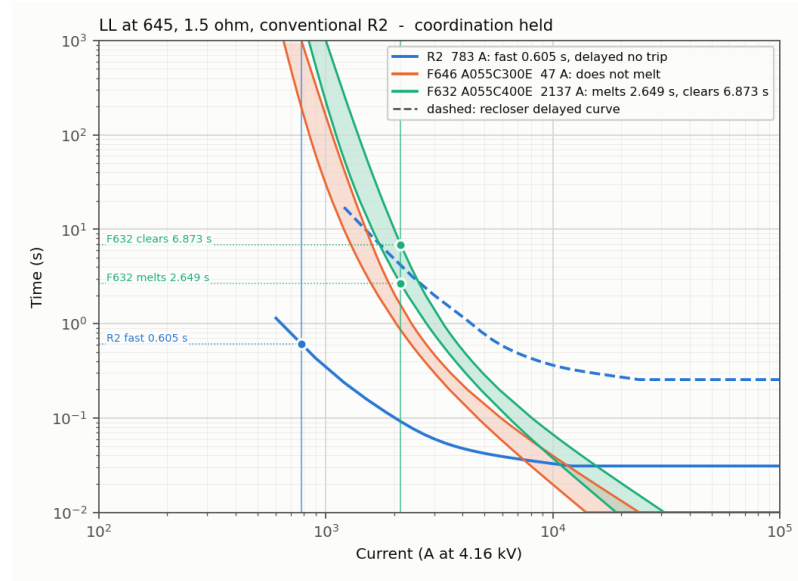


Figure 5.5 LL fault at 645 through 1.5 Ω , conventional R2.

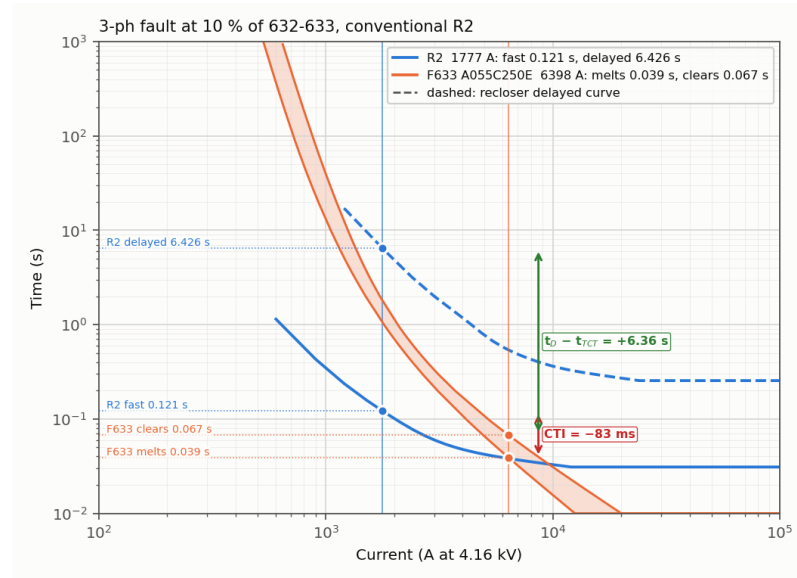


Figure 5.6 Three-phase fault at 10 per cent of line 632–633, conventional R2; CTI = –83 ms, the fuse melts first.

5.8 DSDR Settings and Fuse Revision

The reverse load current of R2 is 252.2 A, so eq. (3.12) gives a reverse pickup of 315.3 A. The reverse group is set on a 500/5 current transformer with plugs of 150 A (fast) and 300 A (delayed) and the same time multipliers 0.1 and 1.0 (Figure 5.7). Step 8 of the method (revising the dials) could not improve any lost case, because R1 is already at both ends of the IAC dial range (0.5 and 10) and R2 at both ends of the CDG table (0.1 and 1.0). Step 9 (revising the fuses) was therefore needed: F632 to 500E; F633, F646, F-DL and

F671-1 to 400E; F692-R to 250E (Table 4.3).

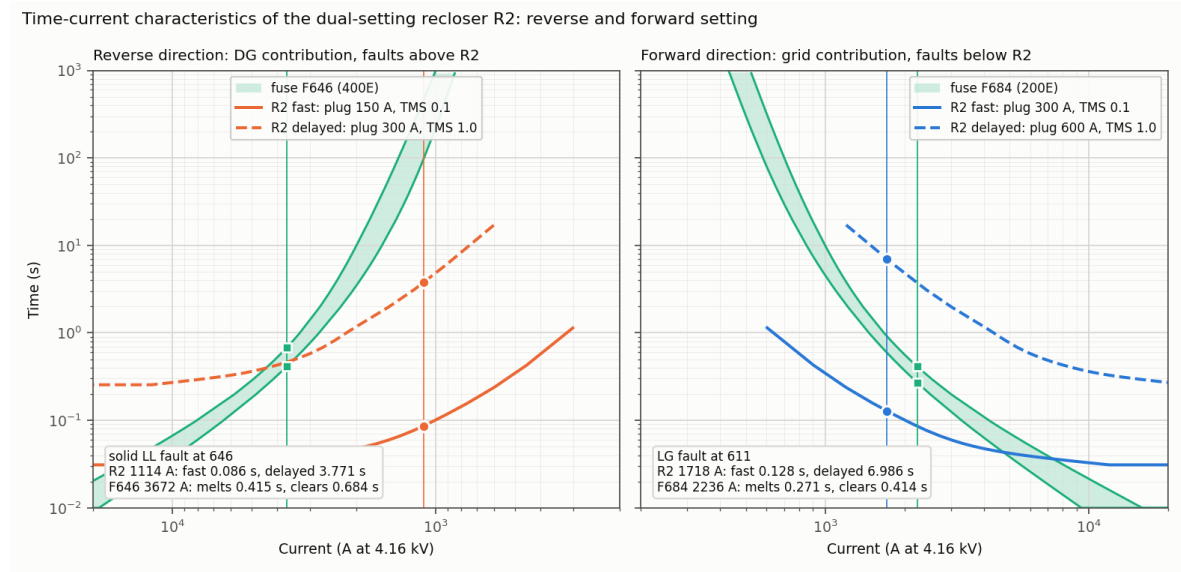


Figure 5.7 The two setting groups of R2: reverse (DG contribution, faults towards 632) and forward (grid contribution, faults beyond 671), each with one fault of the model.

5.9 Coordination With the DSDR

With R2 as a DSDR and the revised fuses, coordination is held in 38 of 39 cells (Table 5.7, Figure 5.8). With the same revised fuses but a single-setting R2 only 31 cells are held. The dual setting itself therefore restores seven cells: all four fault types at 633, and the LG faults at 645, 646 and DL.

The cell that remains lost is the LG fault at 692. F671-1 must be 400E to stay selective with F692-R, but it then clears after R2's delayed trip (7.6 s against 4.0 s), and R2's delayed dial is already at its maximum.

Fault, R2	632	633	645	646	DL	671	692	675	680	684	611	652
LG, single setting	✓	×	×	×	×	✓	×	✓	✓	✓	✓	✓
LG, dual setting	✓	✓	✓	✓	✓	✓	×	✓	✓	✓	✓	✓
LL, single setting	✓	×	✓	✓	✓	✓	✓	✓	✓	✓	—	—
LL, dual setting	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—	—
LLG, single setting	✓	×	✓	✓	✓	✓	✓	✓	✓	✓	—	—
LLG, dual setting	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—	—
LLL, single setting	✓	×	—	—	✓	✓	✓	✓	✓	—	—	—
LLL, dual setting	✓	✓	—	—	✓	✓	✓	✓	✓	—	—	—

Table 5.7 Coordination status with DG and the revised fuses: single-setting R2 against dual-setting R2.

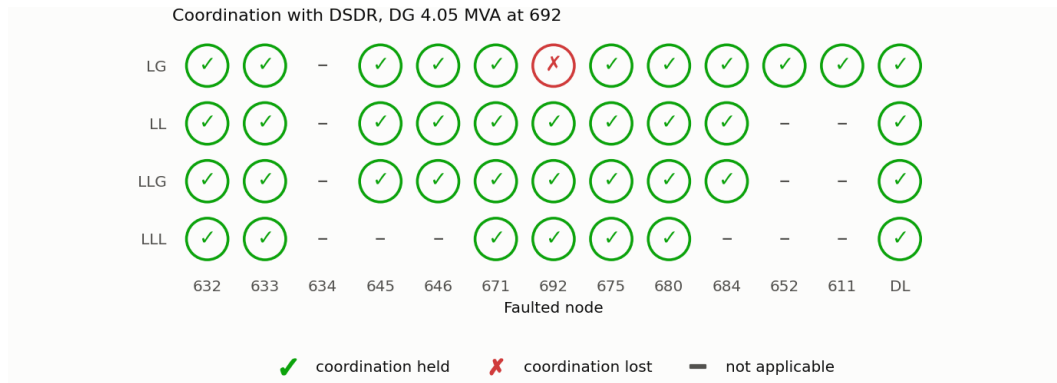


Figure 5.8 Coordination status with DG and the DSDR.

Figures 5.9 and 5.10 show a bolted LL fault at 646. The fuses carry 3.67 kA. R2 carries 1.11 kA in reverse and trips on its reverse fast curve in 0.086 s, before F646 melts (0.415 s).

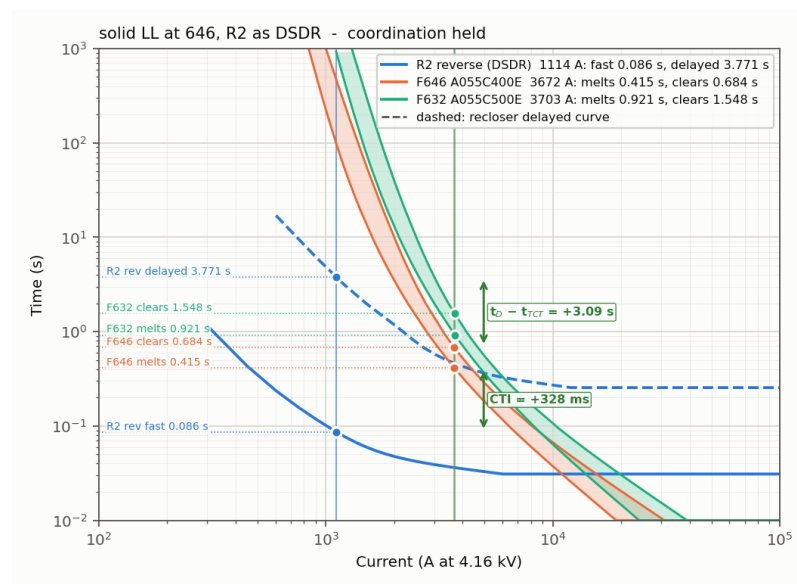


Figure 5.9 Bolted LL fault at 646: R2 reverse setting with F646 and F632; CTI = +328 ms.

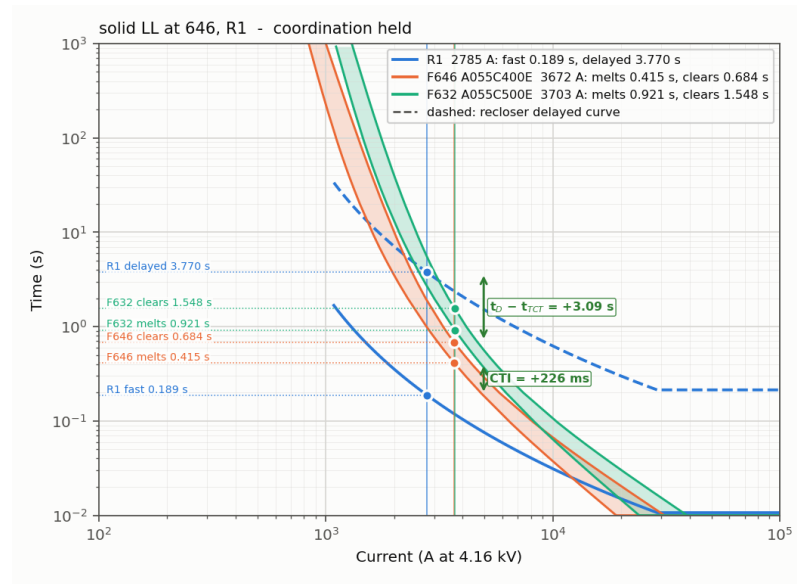


Figure 5.10 Bolted LL fault at 646: R1 with F646 and F632; CTI = +226 ms.

5.10 Operating Times With the DSDR

Tables 5.8 and 5.9 give the operating times with the DSDR. R1 operates forward for every node. R2 operates in reverse for the nodes upstream of it and forward for the nodes downstream. The fuse time is the minimum-melting time of the fuse nearest to the fault at the current through that fuse. In all nine rows with a fuse it is longer than the last fast trip; the smallest margins are 6 ms at 675 and 15 ms at DL and 652.

Node	Fault	I_{R1} (A)	R1 fast (s)	R1 delayed (s)	I_{R2} (A)	Dir.	R2 fast (s)	R2 delayed (s)
632	LLL	4734	0.081	1.627	1813	rev	0.051	1.416
633	LLL	3970	0.106	2.111	1500	rev	0.060	2.001
645	LL	3124	0.155	3.095	1232	rev	0.075	3.002
646	LL	2785	0.189	3.770	1114	rev	0.086	3.771
DL	LLL	3954	0.106	2.123	1972	rev	0.048	1.213
671	LLL	3412	0.134	2.677	3391	fwd	0.054	1.600
692	LLL	3412	0.134	2.677	3391	fwd	0.054	1.600
675	LLL	3090	0.158	3.152	3050	fwd	0.059	1.941
684	LL	2543	0.222	4.437	2543	fwd	0.072	2.813
680	LLL	2789	0.188	3.761	2757	fwd	0.066	2.381
611	LG	1814	0.435	8.709	1718	fwd	0.128	6.986
652	LG	1795	0.446	8.919	1761	fwd	0.123	6.570

Table 5.8 Recloser operating times with the DSDR, DG in service.

Node	Fault	Fuse	Size	I_{fuse} (A)	t_{MMT} (s)	t_{TCT} (s)	Last fast trip (s)	Margin (s)	Status
633	LLL	F633	400E	5426	0.149	0.235	0.106	+0.043	held
645	LL	F632	500E	4191	0.613	0.970	0.155	+0.458	held
646	LL	F646	400E	3672	0.415	0.684	0.189	+0.226	held
DL	LLL	F-DL	400E	5902	0.121	0.193	0.106	+0.015	held
692	LLL	F671-1	400E	3391	0.525	0.891	0.054	+0.471	held
675	LLL	F692-R	250E	5022	0.065	0.106	0.059	+0.006	held
684	LL	F671-2	300E	3963	0.148	0.230	0.072	+0.076	held
611	LG	F684	200E	2236	0.271	0.414	0.128	+0.143	held
652	LG	F652	150E	2239	0.138	0.193	0.123	+0.015	held

Table 5.9 Fuse times with the DSDR, DG in service. Margin = t_{MMT} minus the last fast trip (R2 for faults below R2; the later of R1 and R2 reverse above it). Nodes 632, 671 and 680 are on the main feeder and have no fuse.

5.11 Single Setting Against Dual Setting on Selected Faults

Table 5.10 compares the two settings of R2 on nine faults with the same (revised) fuses: the six faults of the figures above and three cells of the classification. Figures 5.11 and 5.12 show two of the diagrams.

#	Figure	Fault	Z_f (Ω)	I_{R1} (A)	I_{R2} (A), direction	R1 fast (s)	R2 fast (s)		Fuse: melts (s)	Fuse saving	
							single	dual		single	dual
01	Fig. 8	LG fault at 611	0	1814	1718 (fwd)	0.435	0.128	0.128	F684 200E: 0.271	held	held
02	Fig. 9	LLG fault at the middle of 692-675	1	1445	1310 (fwd)	0.742	0.204	0.204	F692-R 250E: 0.618	held	held
03	Fig. 11	LL fault at 646	1	1899	847 (rev)	0.395	0.500	0.131	F646 400E: 1.734	lost	lost
04	Fig. 12	LL fault at 645	1.5	1694	783 (rev)	0.507	0.605	0.149	F632 500E: 9.129	lost	lost
05	Fig. 13	Three-phase fault at 10 % of 632-633	0	4645	1777 (rev)	0.084	0.121	0.052	F633 400E: 0.100	lost	held
06	Figs. 15, 16	LL fault at 646	0	2785	1114 (rev)	0.189	0.278	0.086	F646 400E: 0.415	held	held
07	Fig. 14 / 17 cell	LG fault at 646 (phase c)	0	2359	469 (rev)	0.255	no trip	0.396	F646 400E: 1.057	lost	held
08	Fig. 14 / 17 cell	LG fault at 633 (phase b)	0	2874	644 (rev)	0.178	0.974	0.210	F633 400E: 0.502	lost	held
09	Fig. 14 / 17 cell	LG fault at 692 (phase a)	0	2197	2182 (fwd)	0.293	0.089	0.089	F671-1 400E: 2.842	lost	lost

Table 5.10 Nine faults with the DG in service: single-setting against dual-setting R2, same fuses.

In cases 5, 7 and 8 the dual setting restores the fuse saving: the reverse group picks up a DG current that the forward group does not see (case 7: 469 A) or sees only slowly.

Cases 3, 4 and 9 are lost with both settings; they are limits of the fuse revision and of the dial range, not of the direction. Cases 3 and 4 are faults through impedance: the larger fuses that are needed for the bolted faults are too slow for these lower currents. Cases 5 and 7 are shown in detail in Figures 5.11 and 5.12; Tables 5.11 and 5.12 list their currents, settings and operating times.

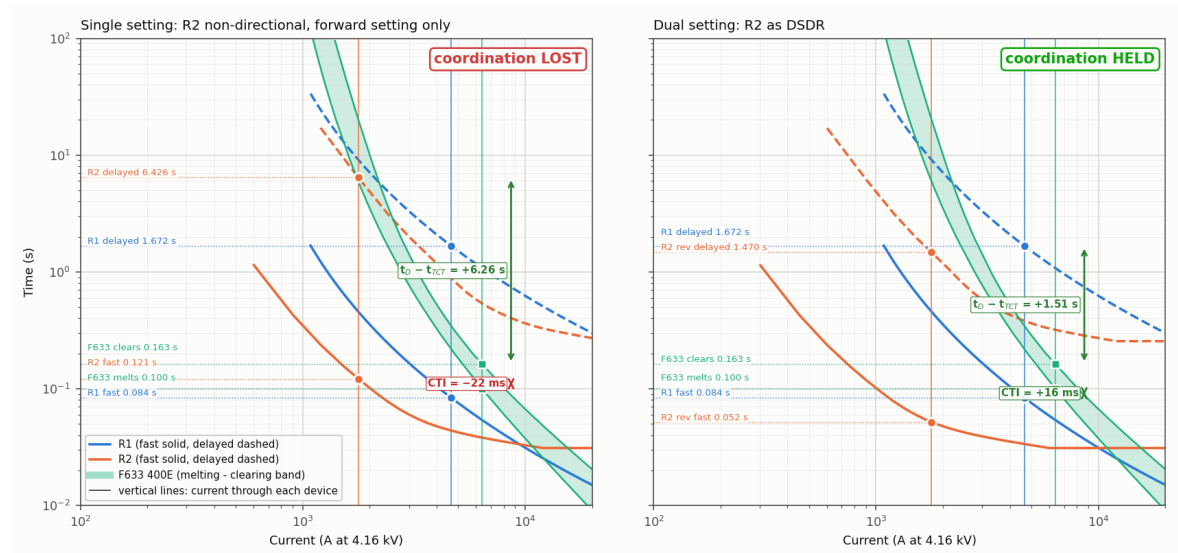


Figure 5.11 Three-phase fault at 10 per cent of line 632–633: single setting (left, CTI = –22 ms) and dual setting (right, CTI = +16 ms).

Quantity	Single setting	Dual setting (DSDR)
Fault	Three-phase fault at 10 % of 632-633, LLL, phases a-b-c	
Fault impedance Z_f	0 Ω (bolted)	
Fault-path current (F633)	6398 A	
Current through R1, direction	4645 A, forward (RG60 → 632), grid contribution	
Current through R2, direction	1777 A, REVERSE (671 → 632), DG contribution only	
R2 setting applied (fast / delayed)	FORWARD: plug 300/600 A, starts 600/1200 A, TMS 0.1/1.0	REVERSE: plug 150/300 A, starts 300/600 A, TMS 0.1/1.0
R1 fast / delayed	0.084 s / 1.672 s	0.084 s / 1.672 s
R2 fast / delayed	0.121 s / 6.426 s	0.052 s / 1.470 s
Primary fuse melts / clears	F633 400E: 0.100 s / 0.163 s	F633 400E: 0.100 s / 0.163 s
Backup fuse melts	-	-
Fuse saving	LOST: F633 melts (0.100 s) before R2 fast (0.121 s)	HELD

Table 5.11 Three-phase fault at 10 per cent of line 632–633: currents, settings and operating times.

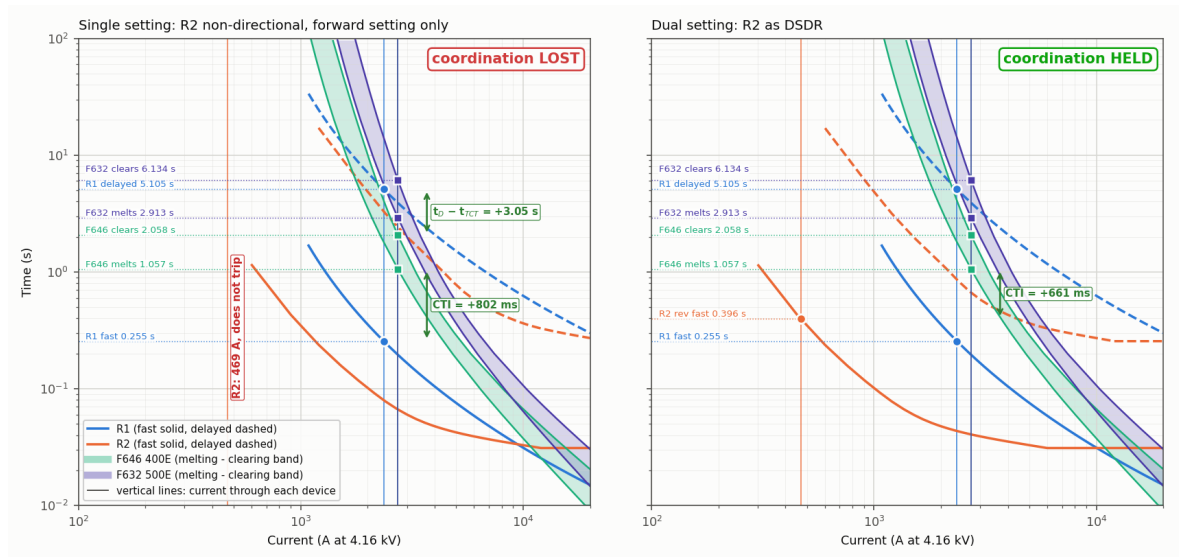


Figure 5.12 LG fault at 646: the single-setting R2 does not pick up the reverse current (left); the reverse group does (right, CTI = +661 ms).

Quantity	Single setting	Dual setting (DSDR)
Fault	LG fault at 646 (phase c), LG, phases C	
Fault impedance Z_f	0 Ω (bolted)	
Fault-path current (F646)	2732 A	
Current through R1, direction	2359 A, forward (RG60 $\rightarrow$ 632), grid contribution	
Current through R2, direction	469 A, REVERSE (671 $\rightarrow$ 632), DG contribution only	
R2 setting applied (fast / delayed)	FORWARD: plug 300/600 A, starts 600/1200 A, TMS 0.1/1.0	REVERSE: plug 150/300 A, starts 300/600 A, TMS 0.1/1.0
R1 fast / delayed	0.255 s / 5.105 s	0.255 s / 5.105 s
R2 fast / delayed	no trip / no trip	0.396 s / no trip
Primary fuse melts / clears	F646 400E: 1.057 s / 2.058 s	F646 400E: 1.057 s / 2.058 s
Backup fuse melts	F632 500E: 2.913 s	F632 500E: 2.913 s
Fuse saving	LOST: R2 does not pick up (469 A)	HELD

Table 5.12 LG fault at 646 (phase c): currents, settings and operating times.

5.12 Time-Sequence Check

The classification above uses the strict rule that every recloser must trip before the fuse starts to melt. A time-sequence check follows each fault in time instead: the fuse heats with the current that still flows after each recloser opens. For the 18 cells upstream of R2 the strict rule gives 11 cells held with the single setting and 18 with the dual setting; the time sequence gives 15 and 18 (Figure 5.13). The three cells that the DSDR restores in the time sequence as well are the LG faults at 633, 645 and 646, where the single-setting R2 never

trips on a reverse current of about 0.5 to 0.6 kA. For the 21 cells downstream of R2 the DG at 692 feeds the fault directly, with no recloser in its path, and the result is the same for both settings.

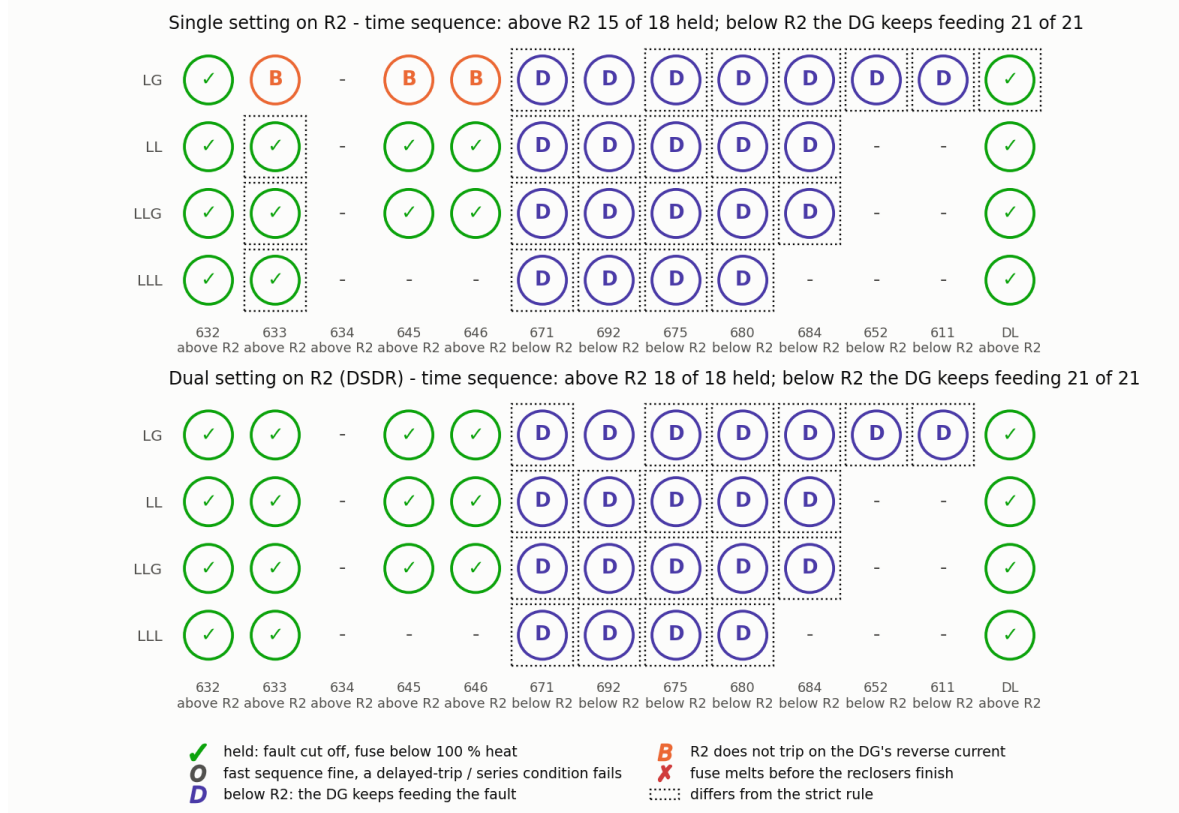


Figure 5.13 Time-sequence check of all 39 cells, single against dual setting.

5.13 Time-Domain Verification

Figure 5.14 is the EMT simulation of an LL (a–c) fault at 684 through 0.2Ω with the conventional settings. The recloser trips follow R2's fast curve applied to the simulated current, and the fuse F671-2 (300E) melts when the accumulated heating $\sum \Delta t / t_{MMT}(I)$ reaches one. Without DG the two fast shots use 45 per cent of the melting heat, the fuse survives both and then clears the permanent fault. With the DG the fuse melts at the end of the second fast shot, because the DG keeps feeding the fault while R2 is open (Table 5.13). The feeder current at node 632 is 557 / 588 A RMS (phases a / c) before the fault and 2362 / 2719 A during it, with peaks of 3338 / 3842 A.

Current at node 632, LL a-c fault at 684 through 0.2 ohm, PowerFactory EMT

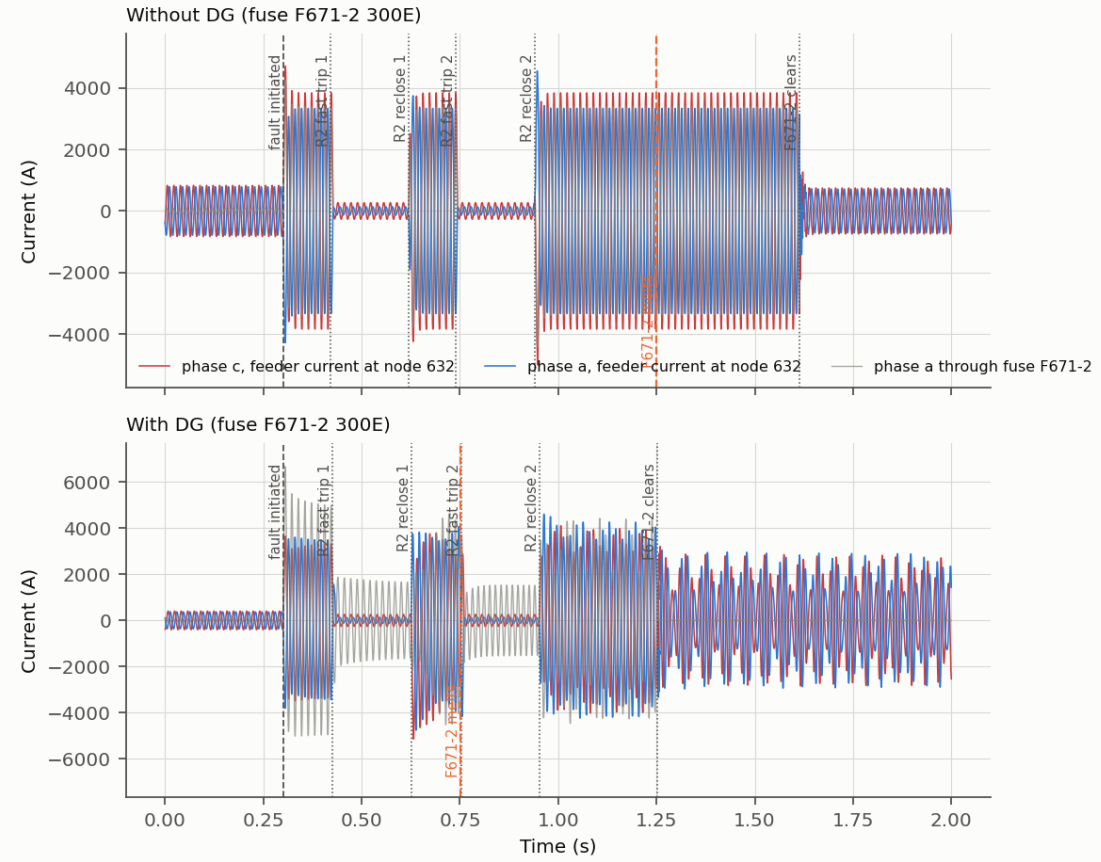


Figure 5.14 Time-domain current at node 632 for an LL (a–c) fault at 684 through 0.2Ω : without DG (top) and with DG (bottom).

Event (time in s)	DG out	DG in
Fault initiated	0.30	0.30
R2 fast trip 1	0.420	0.426
R2 reclose 1	0.620	0.626
R2 fast trip 2	0.740	0.753
R2 reclose 2	0.940	0.953
Fuse F671-2 melts	1.248	0.752
Fuse operation / clears	1.613	1.253
Fuse heat after the fast shots	45 %	105 %

Table 5.13 Reclosing sequence of the EMT simulation.

5.14 Verification With the Relay and Fuse Models of PowerFactory

The operating times of Chapter Five are calculated from the time–current curves of the devices. As a check, the settings were entered into the relay and fuse elements of the model, and PowerFactory itself calculated the tripping time of every device for every fault, for the conventional and for the DSDR settings. Over 1896 operating times the largest deviation is 1.22 per cent.

5.15 Sensitivity of the Result to the Margin

The 38 held cells are obtained with a zero margin. If the breaker interrupting time of three cycles is added to the fast trip, 31 cells are held; if in addition a safety margin is kept on the fuse melting time, 26 cells are held. The pickups are a second limit: with the DG, the lowest LG fault current through $3\ \Omega$ is 577 A at R1 and 368 A at R2, below the pickups of 734.6 A and 587.5 A, because the DG supplies part of the fault.

5.16 Effect of the DG Penetration Level

To see how the conventional coordination degrades as the DG grows, seven separate PowerFactory projects were made, copies of the model with the DG at 692 set to 0, 10, 25, 37, 50, 75 and 100 per cent of 4.05 MVA. The machine, its step-up transformer and its active and reactive power are scaled together, so the per-unit data of Table 4.1 stay the same; at 0 per cent the DG is out of service. Every model keeps the conventional protection of the design without DG (single-setting R2, fuses of the design without DG), and all operating times are PowerFactory’s own relay and fuse calculations. Two fuse–recloser pairs are followed: fuse F633 against R1 for a three-phase fault at 633 (R1’s zone), and the lateral fuse F671-2 at node 671 against R2 for a line-to-line fault at 684 (R2’s zone). The coordination time interval is

$$CTI = t_{MMT}(\text{fuse}) - t_F(\text{recloser}), \quad (5.1)$$

and, for the doughnut chart of Figure 5.15, each level’s share of the ring, $CTI\% = 100 \times CTI / \sum |CTI|$ over the seven levels, so that each ring adds up to 100 per cent.

DG (%)	MVA	Node 633: F633 and R1 (LLL)				Node 671: F671-2 and R2 (LL at 684)			
		I fuse / R1 (A)	t_{MMT} / t_F (s)	CTI (ms)	ring %	I fuse / R2 (A)	t_{MMT} / t_F (s)	CTI (ms)	ring %
0	0.00	4102 / 4162	0.102 / 0.098	+4	+2.1	2574 / 2711	0.470 / 0.067	+403	+26.0
10	0.41	4256 / 4142	0.094 / 0.099	-5	-3.1	2715 / 2681	0.401 / 0.068	+333	+21.5
25	1.01	4478 / 4111	0.083 / 0.100	-17	-9.6	2926 / 2636	0.323 / 0.069	+254	+16.4
37	1.50	4648 / 4088	0.077 / 0.101	-24	-13.9	3094 / 2598	0.277 / 0.070	+207	+13.4
50	2.02	4824 / 4062	0.071 / 0.102	-31	-17.9	3276 / 2557	0.237 / 0.072	+166	+10.7
75	3.04	5140 / 4015	0.062 / 0.104	-42	-24.2	3623 / 2511	0.184 / 0.073	+110	+7.1
100	4.05	5426 / 3970	0.055 / 0.106	-51	-29.2	3963 / 2543	0.147 / 0.072	+75	+4.9

Table 5.14 CTI against DG penetration, conventional protection.

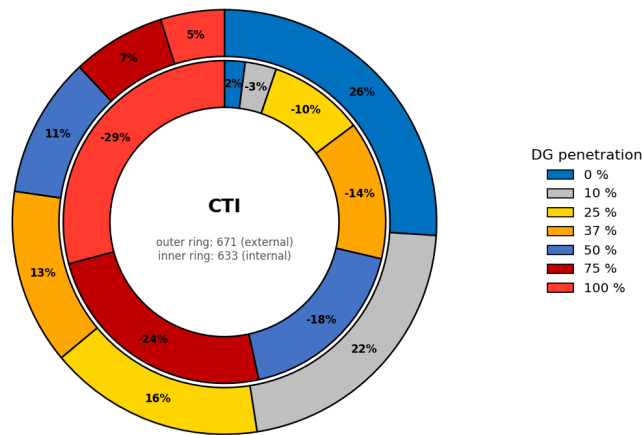


Figure 5.15 Percentage variation of the CTI with DG penetration (outer ring 671, inner ring 633).

At node 633 the fuse current rises from 4102 A to 5426 A while R1's current falls from 4162 A to 3970 A; the CTI falls from +4 ms to -51 ms. At node 671 the fuse current rises from 2574 A to 3963 A while R2's current stays at about 2543 A; the CTI falls from +403 ms to +75 ms. The mechanism is the same at both nodes (Figure 5.16): the fuse lies between the fault and both sources and carries the grid share and the DG share, while the recloser carries only the grid share, which even falls slightly because the DG holds up the voltage of the network during the fault. A fuse curve is steep, so the extra current shortens its melting time much more than the lower current lengthens the recloser's time, and the CTI decreases. At 633 the margin is only a few milliseconds without DG and is lost from about 4 per cent penetration; at 671 the margin is large and stays positive up to 100 per cent.

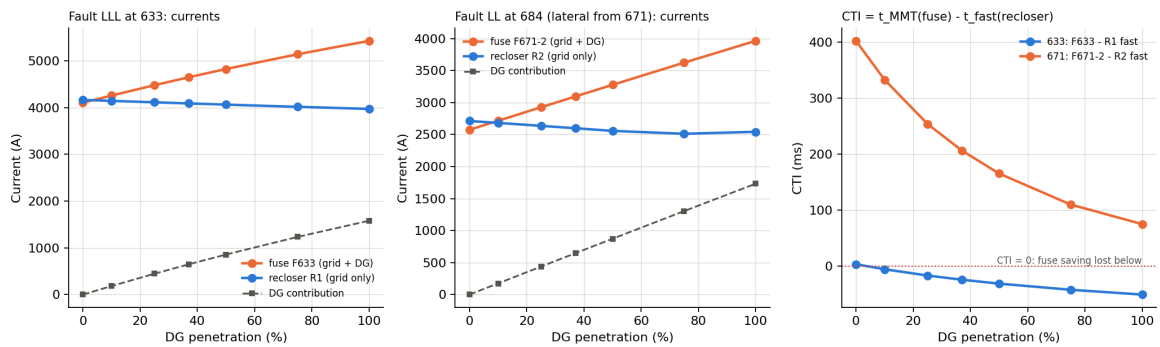


Figure 5.16 Currents through the fuse (grid and DG share), the recloser (grid share) and from the DG, and the resulting CTI, against DG penetration.

Figure 5.17 shows the same on the time–current characteristics: the curves are fixed because the settings are not changed, the recloser’s operating point stays where it is, and the fuse’s operating point moves to a higher current and a shorter time with every step. Figure 5.18 gives node 633 level by level, with the vertical short-circuit lines of the fuse and the recloser currents, the operating times and the CTI.

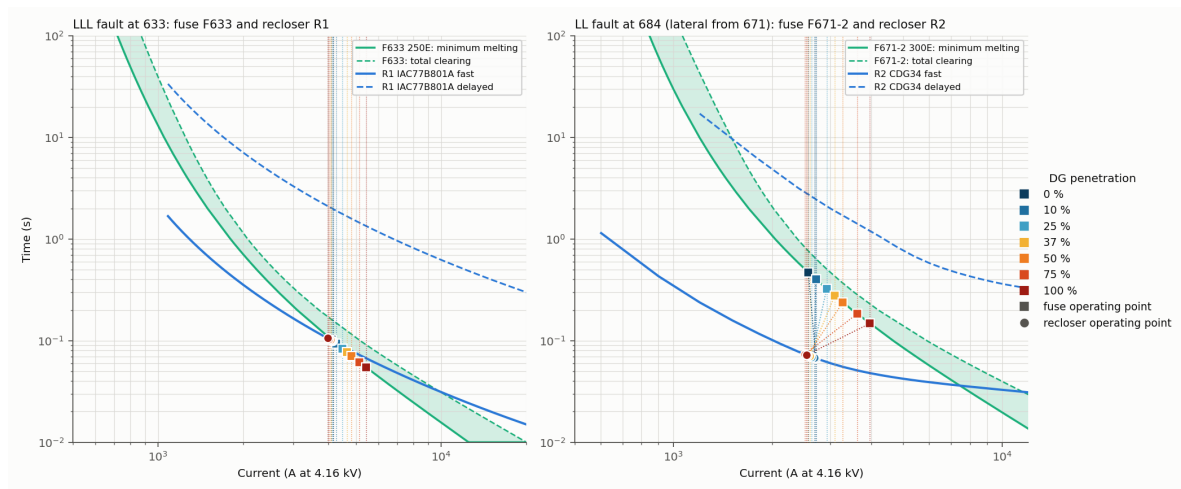


Figure 5.17 Time–current characteristics at all penetration levels: fuse operating points (squares) and recloser operating points (circles).

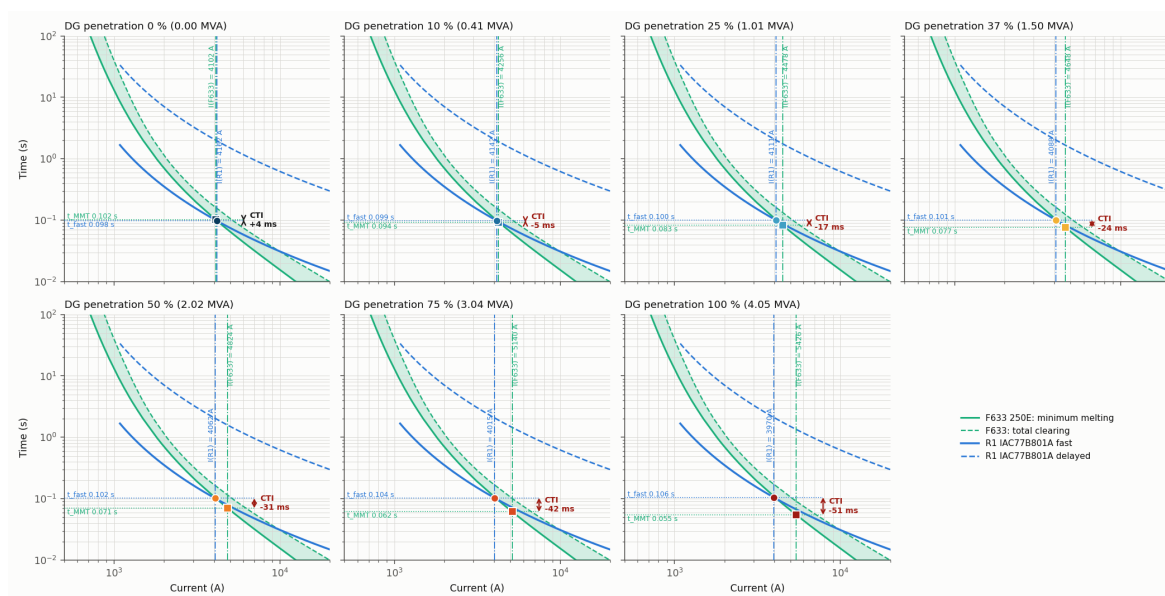


Figure 5.18 Node 633, fuse F633 and R1, at each penetration level, with the short-circuit currents, operating times and CTI.

The fault level rises at every node (Table 5.15), most near the DG (671, 692, 675: about +60 per cent) and least behind the 4.16/0.48 kV transformer (634) and on the single-phase laterals. The current through the reclosers does not rise with it.

Node	Fault	0 %	25 %	50 %	75 %	100 %	Change
632	LLL ABC	4735	5227	5690	6123	6526	+38 %
633	LLL ABC	4102	4478	4824	5140	5426	+32 %
634	LLL ABC	15671	16307	16828	17257	17608	+12 %
645	LLG BC	3418	3685	3932	4149	4341	+27 %
646	LLG BC	3112	3332	3528	3701	3854	+24 %
671	LLL ABC	3390	3919	4461	5010	5564	+64 %
692	LLL ABC	3390	3919	4461	5010	5564	+64 %
675	LLL ABC	3151	3612	4087	4557	5022	+59 %
680	LLL ABC	2930	3322	3711	4091	4460	+52 %
684	LLG AC	2655	3027	3395	3756	4107	+55 %
652	LG A	1874	1983	2080	2164	2239	+19 %
611	LG C	1857	1973	2074	2161	2236	+20 %
DL	LLL ABC	3946	4454	4952	5436	5902	+50 %

Table 5.15 Largest bolted fault current at each node (A) against DG penetration.

The load flow changes as well (Table 5.16). The DG supplies part of the load locally, so the load current through both reclosers falls, the voltages rise, and the power through R2

reverses between 75 and 100 per cent. By eq. (3.3) the pickups would have to fall with the penetration; with the fixed settings (R1 720 A, its curve starting at 1080 A; R2 fast curve starting at 600 A) the reclosers no longer detect the weakest faults (LG through 3 Ω) in their zones, R2 from just above 50 per cent.

DG (%)	DG P (MW)	R1 load (A)	R2 load (A)	R2 flow	$V_{671,a}$ (p.u.)	R1 1.25 I_{nom} (A)	R2 1.25 I_{nom} (A)	R1 lowest 3 Ω fault (A)	R2 lowest 3 Ω fault (A)
0	0.00	587.7	470.0	forward	0.988	734.6	587.5	1074*	878
10	0.32	543.1	430.1	forward	0.994	678.9	537.6	1018*	821
25	0.81	480.7	372.2	forward	1.002	600.9	465.2	937*	738
37	1.20	434.8	328.6	forward	1.009	543.5	410.8	874*	673
50	1.62	389.9	285.3	forward	1.015	487.4	356.6	809*	606
75	2.43	321.2	222.4	forward	1.025	401.5	278.0	689*	483*
100	3.24	284.9	252.2	reverse	1.033	356.1	315.2	577*	368*

Table 5.16 Load flow, pickups by eq. (3.3) and lowest fault current through each recloser against DG penetration. *: below the start of the recloser's curve with the fixed settings (not detected).

The penetration study therefore shows the two ways in which fixed settings fail as the DG grows: the fuse saving is lost because the fuse speeds up while the recloser does not, and the sensitivity is lost because the currents through the reclosers fall. Both are reasons for settings that adapt to the DG, which is what the dual-setting directional recloser provides.

5.17 Results Summary

Stage	Cells held (of 39)
A. No DG, starting fuse sizes	35
A. No DG, after the fuse revision (F692-R to 200E)	39
B. DG connected, conventional R2	24
B. DG connected, no-DG fuses, R2 as DSDR	25
C. DG connected, revised fuses, conventional R2	31
C. DG connected, revised fuses, R2 as DSDR	38
C. The same settings and fuses with the DG out of service	37

Table 5.17 Summary of the coordination results.

The dual setting alone, with the fuses of the design without DG, restores one cell (24 to 25). Together with the fuse revision it restores fourteen (24 to 38), seven of which need the dual setting.

CHAPTER SIX: DISCUSSION

6.1 Effect of DG on Fault Current

The DG raises the fault current at every node, most near its connection point (671 and 692: from 3.39 to 5.56 kA). The fuses carry this higher current and melt faster, while R1 carries about the same grid current as before. The margin between the fuse and the fast trip, which is small even without DG, disappears.

6.2 Effect of Fault Current Direction

For faults at 632, 633, 645, 646 and DL the current through R2 is reverse and consists only of the DG contribution. Its magnitude (0.5 to 2.0 kA) is much lower than the forward fault currents for which the forward group is set. A forward group with plugs of 300 and 600 A responds slowly to these currents, or not at all for LG faults.

6.3 Performance of the Conventional Protection

With the conventional single setting the scheme keeps coordination at the nodes on the main feeder and loses it on the fused laterals of R1's zone and at 675. The loss depends on the location and on the fault type; it is not a uniform failure of the feeder.

6.4 Performance of the DSDR

The reverse group, with half the plug settings, trips for every reverse fault before the fuse melts once the fuses are revised. The result of 38 cells against 31 with the same fuses isolates its contribution. Without the fuse revision the improvement is one cell only: at 633 and DL the fuses of the design without DG melt in 0.05 to 0.16 s with the DG, before R1's fast trip (0.11 to 0.18 s), and no setting of R2 can change R1's time. The DSDR is therefore necessary but not sufficient.

6.5 Limits Reached by the Devices

R1's dials are the two extremes of the IAC range, and the CDG table ends at 0.1 and 1.0. The method's dial revision has no room. The fuse revision creates a conflict between the two states of the network: the larger fuses that are needed with the DG are too slow for two cells without it, and for the faults through impedance. Fuse sizes cannot adapt to the

DG state. For faults downstream of R2 the fuse saving relies on the DG's own protection disconnecting it during the dead time, which is not part of the method.

6.6 Engineering Implication

Adaptive directional protection is justified where a DG is downstream of a mid-line recloser. To apply it, the engineer must check the dial range of the actual relay, revise the fuse sizes for the DG state, verify the pickup against the reduced minimum fault current with DG, and provide for the DG infeed to faults beyond the recloser.

CHAPTER SEVEN: CONCLUSION AND FUTURE WORK

7.1 Conclusion

The DSDR method of [1] was implemented on the IEEE 13-node feeder in DIgSI-LENT PowerFactory with the manufacturer-based time–current characteristics of the protection devices. The feeder model agrees with the IEEE short-circuit benchmark within about 2 per cent. Without DG, the revised fuse scheme achieves coordination in all 39 cells, compared with 35 of 39 cells with the starting fuse sizes. With the 4.05 MVA DG at node 692 and a conventional R2, coordination is held in 24 cells. The dual setting alone, with the fuses of the design without DG, raises this to 25 cells; the fuse revision alone, with a single-setting R2, to 31; and the two together to 38 of 39 cells. The combined DSDR and fuse-revision strategy therefore substantially improves the coordination in the zone between the grid and the DG, achieving 38 of 39 coordinated fault cells under the zero-margin criterion, and seven of the restored cells need the dual setting. The remaining LG fault at node 692 is limited by the maximum available R2 delayed dial.

The result depends on the margin. When a three-cycle breaker interrupting time is added to the fast trip, the number of coordinated cells decreases from 38 to 31, and with an additional fuse safety margin to 26. The faults downstream of R2 with the DG in service also remain outside what the dual setting can correct. The relay and fuse models of PowerFactory confirm the calculated operating times within 1.22 per cent.

7.2 Major Findings

1. Fault levels at the IEEE 13-node benchmark (feeder head 4.73 kA).
2. R1: pickup 720 A, TDS 0.5 / 10, GE IAC extremely inverse curve.
3. R2 forward: plugs 300 / 600 A, TMS 0.1 / 1.0; R2 reverse: plugs 150 / 300 A, TMS 0.1 / 1.0; reverse pickup by eq. (3.12) 315.3 A.
4. With DG the load current at R2 reverses (252 A) and R2 carries 1.1 to 2.0 kA in reverse for upstream faults.
5. Coordination (of 39 cells): 35 without DG with the starting fuses and 39 after the fuse revision; 24 with DG and a conventional R2; 25 with the DSDR alone; 31 with the fuse revision alone; 38 with the DSDR and the fuse revision, under the zero-margin criterion.
6. The 38 cells become 31 when a three-cycle breaker interrupting time is added to the

fast trip, and 26 when a safety margin is also kept on the fuse melting time.

7. The dual setting restores seven cells with the revised fuses and one cell without the revision.
8. The LG fault at 692 cannot be coordinated within the delayed dial range of R2.
9. With rising DG penetration and fixed settings the CTI falls at both studied nodes (633: from about +4 to −51 ms; 671: from +403 to +75 ms), the fault level rises by up to 64 per cent, and R2 stops detecting the weakest fault of its zone above about 50 per cent.
10. PowerFactory’s own relay and fuse models confirm the operating times within 1.22 per cent.
11. In the EMT simulation the fuse survives the two fast shots without DG and melts at 1.25 s; with DG it melts during the second fast shot.

7.3 Contributions of the Present Work

- A reproducible PowerFactory implementation of the method: one command rebuilds the model and all results.
- Separation of the effect of the dual setting from the effect of the fuse revision.
- Identification of the limits of the method: dial range, DG-in against DG-out fuse sizes, DG infeed downstream of the recloser, and sensitivity of the pickups with DG.
- A time-domain verification and a verification with the relay models of the tool.

7.4 Future Work

Future work may include the extension of the study to the IEEE 34-node test feeder, an explicit directional element for R2, a classification with a practical margin (breaker time and a fuse margin), a recloser with a wider dial range or with user-defined curves, a setting-group change triggered by the DG state, the disconnection logic of the DG during the reclosing dead time, the sensitivity to fault impedance, inverter-based generation, and hardware-in-the-loop tests.

REFERENCES

- [1] M. Yousaf, A. Jalilian, K. M. Muttaqi, and D. Sutanto, “An adaptive overcurrent protection scheme for dual-setting directional recloser and fuse coordination in unbalanced distribution networks with distributed generation,” *IEEE Transactions on Industry Applications*, vol. 58, no. 2, pp. 1831–1842, Mar./Apr. 2022, doi: 10.1109/TIA.2022.3146095.
- [2] G. Benmouyal *et al.*, “IEEE standard inverse-time characteristic equations for overcurrent relays,” *IEEE Transactions on Power Delivery*, vol. 14, no. 3, pp. 868–872, Jul. 1999.
- [3] A. Naiem, Y. Hegazy, A. Abdelaziz, and M. Elsharkawy, “A classification technique for recloser-fuse coordination in distribution systems with distributed generation,” *IEEE Transactions on Power Delivery*, vol. 27, no. 1, pp. 176–185, Jan. 2012.
- [4] GE Multilin, *735/737 Feeder Protection Relay – Instruction Manual*, manual P/N 1601-0048-DK (GEK-106291F), Markham, ON, Canada, 2010, Ch. 5: Overcurrent Curves (IAC curves).
- [5] W. H. Kersting, “Radial distribution test feeders,” *IEEE Transactions on Power Systems*, vol. 6, no. 3, pp. 975–985, Aug. 1991.
- [6] M. Yousaf, K. M. Muttaqi, and D. Sutanto, “Overcurrent protection scheme for the IEEE 13-node benchmark test feeder with improved selectivity,” in *Proc. IEEE Power & Energy Society General Meeting*, 2020, doi: 10.1109/PESGM41954.2020.9281529.
- [7] W. H. Kersting and G. Shirek, “Short circuit analysis of IEEE test feeders,” in *Proc. IEEE PES Transmission and Distribution Conference and Exposition*, 2012.
- [8] DIgSILENT GmbH, *PowerFactory 2021 User Manual*, Gomaringen, Germany, 2021.

APPENDIX A: COMPLETE COPY OF THE REFERENCE PAPER

The complete reference paper [1] is attached as a separate, unchanged PDF copy in accordance with the requirement of the department. It is not reproduced inside this document.

APPENDIX B: IMPLEMENTATION FORMULAS

The operating times are evaluated with the following expressions, where $M = I_f/I_p$ for R1 and $M = I_f/I_s$ for R2.

$$\text{R1 (IAC): } t = TDS \left[0.004 + \frac{0.6379}{M - 0.62} + \frac{1.7872}{(M - 0.62)^2} + \frac{0.2461}{(M - 0.62)^3} \right] \quad (\text{B.1})$$

$$\text{R2 (CDG table): } t = t_1 \left(\frac{t_2}{t_1} \right)^f, \quad f = \frac{\ln(M/M_1)}{\ln(M_2/M_1)} \quad (\text{B.2})$$

$$\text{Fuse (curve points): } t = t_1 \left(\frac{t_2}{t_1} \right)^f, \quad f = \frac{\ln(I/I_1)}{\ln(I_2/I_1)} \quad (\text{B.3})$$

$$\text{Fuse saving: } t_{MMT}(I_{fuse}) > t_F, \quad t_{TCT}(I_{fuse}) < t_D \quad (\text{B.4})$$

$$\text{Series fuses: } t_{TCT,i} \leq 0.75 t_{MMT,i+1} \quad (\text{B.5})$$

$$\text{Fuse heating (EMT): } \sum_j \frac{\Delta t_j}{t_{MMT}(I_j)} = 1 \Rightarrow \text{the fuse melts} \quad (\text{B.6})$$

Worked example, node 633, three-phase fault, DG in service. R1 carries 3970 A: $M = 3970/720 = 5.514$, $M - 0.62 = 4.894$, bracket $= 0.004 + 0.1303 + 0.0746 + 0.0021 = 0.2111$, so $t_F = 0.5 \times 0.2111 = 0.106$ s and $t_D = 10 \times 0.2111 = 2.111$ s. R2 carries 1500 A in reverse: fast, $M = 1500/150 = 10.0$, table 0.060 s; delayed, $M = 1500/300 = 5.0$, table 2.0 s. The fuse F633 (400E) carries 5426 A, the sum of both contributions: its curve points are 4800 A at 0.2 s and 6392 A at 0.1 s, $f = \ln(5426/4800)/\ln(6392/4800) = 0.428$, $t_{MMT} = 0.2 \times 0.5^{0.428} = 0.149$ s. The fuse outlasts the last fast trip (0.106 s) by 0.043 s.

APPENDIX C: COMPARISON WITH THE REFERENCE PAPER

This appendix compares the results of Chapter Five with the values printed in the reference paper [1] or read from its figures.

Basis of Comparison

The comparison uses only values that the paper prints or that can be read from its figures. The model was not tuned to match the paper; differences are stated and explained from the evidence in the model.

Aspect	This project	Reference paper	Assessment
Rated branch currents	within 2 %	Table II	Very good
Fault levels	IEEE benchmark, feeder head 4.73 kA	5.41 kA, some branches higher	Lower
R1 worked example (4219 A)	0.096 / 1.926 s	0.097 / 1.932 s	Very good
Recloser curves	IAC77B801A and CDG34 library curves	Same devices	Same
Fuse coefficients, series fuses of R1's zone	within 0.05 with reversed index	Table III	Index reversed in the paper
DG model	Dynamic synchronous machine, Table I data	Synchronous machine	Same
Coordination with DG, conventional R2	24 of 39 held; 29 cells agree	30 of 39 held (Fig. 14)	Good
Coordination with the DSDR	38 of 39 held	39 of 39 held (Fig. 17)	Good; one cell at the dial limit
Fuse current, LL at 646	3.67 kA	3.64 kA (Fig. 15)	Very good
R2 reverse current, LL at 646	1.11 kA	1.54 kA	Lower
Table IV, R1 times	12 to 78 % slower (652: 34 % faster)	Table IV	Follows the fault levels
Table IV, R2 and fuse times	assumed R2 settings, revised fuses	settings not published	Not comparable in detail
EMT sequence, fuse operation	melts 1.25 s, clears 1.61 s	1.21 s (Fig. 10)	Close if 1.21 s is melting
IEEE 34-node feeder	not modelled	validated	Remaining work

Table C.1 Comparison of the results with the reference paper.

Base-Case Comparison

The rated currents test the feeder model independently of the protection and of the DG. Their agreement within 2 per cent shows that the topology, the loads and the regulator are implemented as in the paper.

Fault-Current Magnitude Difference

The maximum fault currents of the feeder branches are 4 to 60 per cent below the paper's Table II. The model was validated against the published IEEE short-circuit results instead, which it meets within about 2 per cent. The paper's values cannot all be correct: 6.73 kA on 632–633 and 7.83 kA on 692–675 exceed its own 5.41 kA at the feeder head. The two worked cases of the paper illustrate the remaining difference: R2 carries 1718 A for the LG fault at 611 (paper: 1974 A) and 1114 A in reverse for the LL fault at 646 (paper: 1542 A).

Fuse and Coordination Comparison

The paper's Table III could be reproduced only with the series index reversed. The fuse times in the paper's figures do not follow its fuse equation: for F646 at 3.6 kA the figure shows about 0.18 s, the equation with Table III gives 1.85 s. The figures match A055C library fuses, and the sizes change between figures (F646 is 200E in Fig. 12 and 300E in Figs. 11 and 16). The model therefore uses library fuse curves throughout.

Operating-Time Comparison (Table IV)

The R1 equation and settings reproduce the paper's worked example, so the difference of the R1 times comes from the fault currents. In every row of the paper the delayed time of R1 is 20 times the fast time, the ratio of the dials 10 and 0.5, which confirms the dials. For R2 the paper prints no settings and the ratio of its two times is not constant, so the R2 times and the fuse times of the model are the result of the stated method and cannot be matched to the paper row by row.

Central Finding Comparison

The central finding of the paper is reproduced: the DG makes a conventional scheme lose coordination in the zone between the grid and the DG, and a dual-setting directional R2 restores it there. The model adds two qualifications. The dual setting needs the fuse revision

to be effective; and it does not help for faults downstream of R2, which the DG feeds directly.

Tables of the comparison

Parameter	Symbol	Paper	Model	Unit
Leakage reactance	X_l	0.05	0.05	pu
Stator resistance	R_a	0.0014	0.0014	pu
d-axis synchronous reactance	X_d	1.4	1.4	pu
d-axis transient reactance	X'_d	0.231	0.231	pu
d-axis subtransient reactance	X''_d	0.118	0.118	pu
d-axis transient open-circuit time constant	T'_{d0}	5.5	5.5	s
d-axis subtransient open-circuit time constant	T''_{d0}	0.05	0.05	s
q-axis synchronous reactance	X_q	1.372	1.372	pu
q-axis transient reactance	X'_q	0.8	0.8	pu
q-axis subtransient reactance	X''_q	0.118	0.118	pu
q-axis transient open-circuit time constant	T'_{q0}	1.25	1.25	s
q-axis subtransient open-circuit time constant	T''_{q0}	0.19	0.19	s
Mechanical starting time	$M = 2H$	1.5	1.5	s
DG rating	S_n	4.05	4.05	MVA
DG voltage	U_n	0.69	0.69	kV
Step-up transformer, HV side	U_{HV}	4.16	4.16	kV
Step-up transformer leakage reactance	x_T	0.15	0.15	pu

Table C.2 DG parameters: reference paper (Table I) against the PowerFactory model.

Branch	I_{nom} (A)			$I_{f,min}$ (kA)		$I_{f,max}$ (kA)		
	model	paper	dev. %	model	paper	model	paper	dev. %
RG60–632	587.7	587.1	+0.1	1.07	1.24	4.73	5.41	–13
632–633	81.0	81.1	–0.1	0.76	0.72	4.1	6.73	–39
632–645	143.1	143.3	–0.1	0.73	0.93	3.42	4.25	–20
632–671	470.0	478.2	–1.7	0.88	1.08	3.39	4.52	–25
XFM1-HV side	81.0	81.1	–0.1	0.08	0.73	1.81	6.48	–72
XFM1-LV side	706.4	704.7	+0.2	0.71	0.59	15.67	18.76	–16
645–646	66.1	64.9	+1.8	0.73	0.84	3.11	3.64	–15
671–692	229.7	229.3	+0.2	0.74	0.78	3.39	3.69	–8
671–684	71.1	70.9	+0.3	0.66	0.69	2.66	3.47	–23
671–680	0.0	0.0	–	0.64	0.59	2.93	6.25	–53
692–675	205.9	205.9	+0.0	0.74	0.71	3.15	7.83	–60
684–611	71.1	70.8	+0.4	0.68	0.63	1.86	1.93	–4
684–652	63.0	63.1	–0.2	0.66	0.66	1.87	2.67	–30

Table C.3 Branch currents: model against Table II of the paper. XFM1 HV side: the model values are 4.16 kV currents for a fault on the 0.48 kV side, so this row is not comparable.

The maximum fault currents of the model are lower than those of the paper's Table II. To decide which values are right, both are compared in Table C.4 with the IEEE 13-node short-circuit benchmark [7], as quoted by the same authors in [6]. The model agrees with the benchmark within 2 per cent at 10 of the 11 nodes (largest difference 4.7 per cent), while the paper's values lie 4 to 151 per cent above it. Two of the paper's values, 6.73 kA on 632–633 and 7.83 kA on 692–675, even exceed its own 5.41 kA at the feeder head, which is not possible in a radial feeder without DG. The lower fault levels of this study are therefore those of the IEEE benchmark, not an error of the model.

Node	IEEE benchmark	This study		Reference paper	
	(kA)	If,max (kA)	dev. %	If,max of branch (kA)	dev. %
632	4.80	4.74	−1.3	5.41	+12.7
633	4.15	4.10	−1.1	6.73	+62.2
645	3.41	3.42	+0.2	4.25	+24.6
646	3.05	3.11	+2.0	3.64	+19.3
671	3.35	3.39	+1.2	4.52	+34.9
692	3.35	3.39	+1.2	3.69	+10.1
675	3.12	3.15	+1.0	7.83	+151.0
680	2.91	2.93	+0.7	6.25	+114.8
684	2.62	2.66	+1.3	3.47	+32.4
652	1.79	1.87	+4.7	2.67	+49.2
611	1.85	1.86	+0.4	1.93	+4.3

Table C.4 Maximum fault current, DG out: IEEE benchmark, this study and the paper’s Table II. Deviations are relative to the benchmark.

Fuse	I_f (A)	i/z	t_F (s)	t_D (s)	t_{fuse} (s)	b_i	b_i rev.	paper	Size	t_{MMT} (s)
F632	3418	2/2	0.133	2.670	1.824	6.62	6.35	6.38	400E	0.512
F633	4102	1/1	0.100	2.008	1.054	6.53	6.53	6.83	250E	0.102
F634	15671	1/2	0.439	8.772	3.217	8.06	8.33	8.32	gL-800A	0.021
F645	3418	1/2	0.133	2.670	0.979	6.35	6.62	6.65	250E	0.158
F646	3112	1/2	0.156	3.115	1.142	6.35	6.62	6.67	300E	0.273
F-DL	3946	1/1	0.106	2.130	1.118	6.52	6.52	6.54	250E	0.112
F671	3390	1/1	0.054	1.601	0.827	6.27	6.27	6.08	250E	0.161
F671-1	3390	3/3	0.054	1.601	1.214	6.44	6.00	5.85	300E	0.218
F692	3390	1/2	0.054	1.601	0.569	6.11	6.39	6.13	200E	0.094
F692-R	3151	2/3	0.057	1.829	0.943	6.27	6.27	6.14	200E	0.112
F675	3151	1/3	0.057	1.829	0.500	6.00	6.44	6.19	200E	0.112
F671-2	2655	3/3	0.069	2.572	1.946	6.45	6.01	5.87	300E	0.428
F684	1857	2/3	0.113	5.778	2.946	6.35	6.35	6.15	200E	0.465
F611	1857	1/3	0.113	5.778	1.530	6.07	6.52	6.19	125E	0.120
F652	1874	1/2	0.112	5.660	1.961	6.18	6.47	6.17	150E	0.207

Table C.5 Fuse coefficients against Table III of the paper. “ b_i rev.”: index i counted from the source, which reproduces the paper’s values in R1’s zone.

Node	Fault	I_{R1} (A)	R1 fast (s) model / paper	R1 delayed (s) model / paper	I_{R2} (A)	Dir.	R2 fast (s) model / paper	R2 delayed (s) model / paper
632	LLL	4734	0.081 / 0.070	1.627 / 1.398	1813	rev	0.051 / 0.045	1.416 / 0.514
633	LLL	3970	0.106 / 0.094	2.111 / 1.880	1500	rev	0.060 / 0.052	2.001 / 0.706
645	LL	3124	0.155 / 0.105	3.095 / 2.095	1232	rev	0.075 / 0.087	3.002 / 1.659
646	LL	2785	0.189 / 0.128	3.770 / 2.555	1114	rev	0.086 / 0.103	3.771 / 2.046
DL	LLL	3954	0.106 / 0.087	2.123 / 1.733	1972	rev	0.048 / 0.076	1.213 / 1.402
671	LLL	3412	0.134 / 0.105	2.677 / 2.091	3391	fwd	0.054 / 0.068	1.600 / 1.113
692	LLL	3412	0.134 / 0.115	2.677 / 2.307	3391	fwd	0.054 / 0.072	1.600 / 1.258
675	LLL	3090	0.158 / 0.121	3.152 / 2.414	3050	fwd	0.059 / 0.074	1.941 / 1.332
684	LL	2543	0.222 / 0.125	4.437 / 2.507	2543	fwd	0.072 / 0.075	2.813 / 1.394
680	LLL	2789	0.188 / 0.137	3.761 / 2.747	2757	fwd	0.066 / 0.079	2.381 / 1.555
611	LG	1814	0.435 / 0.326	8.709 / 6.523	1718	fwd	0.128 / 0.146	6.986 / 4.385
652	LG	1795	0.446 / 0.676	8.919 / 13.525	1761	fwd	0.123 / 0.143	6.570 / 9.370

Table C.6 Recloser operating times against Table IV of the paper.

Node	Fault	Fuse	Size	I_{fuse} (A)	t_{MMT} (s)		t_{TCT} (s)	Last fast trip (s)	Margin (s)	Status
633	LLL	F633	400E	5426	0.149	0.163	0.235	0.106	+0.043	held
645	LL	F632	500E	4191	0.613	0.276	0.970	0.155	+0.458	held
646	LL	F646	400E	3672	0.415	0.181	0.684	0.189	+0.226	held
DL	LLL	F-DL	400E	5902	0.121	0.248	0.193	0.106	+0.015	held
692	LLL	F671-1	400E	3391	0.525	0.175	0.891	0.054	+0.471	held
675	LLL	F692-R	250E	5022	0.065	0.185	0.106	0.059	+0.006	held
684	LL	F671-2	300E	3963	0.148	0.205	0.230	0.072	+0.076	held
611	LG	F684	200E	2236	0.271	0.417	0.414	0.128	+0.143	held
652	LG	F652	150E	2239	0.138	0.233	0.193	0.123	+0.015	held

Table C.7 Fuse times against Table IV of the paper.

Fault	632	633	645	646	DL	671	692	675	680	684	611	652
LG (model)	✓	×	×	×	×	✓	✓	✓	✓	✓	✓	✓
LG (paper)	✓	✓	×	✓	✓	✓	✓	✓	✓	✓	✓	✓
LL (model)	✓	×	✓	×	×	✓	✓	×	✓	✓	–	–
LL (paper)	✓	✓	×	×	✓	✓	✓	✓	✓	✓	–	–
LLG (model)	✓	×	✓	×	×	✓	✓	×	✓	✓	–	–
LLG (paper)	✓	×	×	×	✓	✓	✓	×	✓	✓	–	–
LLL (model)	✓	×	–	–	×	✓	✓	×	✓	–	–	–
LLL (paper)	✓	×	–	–	✓	✓	✓	×	✓	–	–	–

Table C.8 Coordination status with DG and a conventional R2: model against the paper's Fig. 14.

Event (time in s)	Paper (read off Fig. 10)	Model, DG out	Model, DG in
Fault initiated	0.30	0.30	0.30
R2 fast trip 1	≈0.46	0.420	0.426
R2 reclose 1	≈0.65	0.620	0.626
R2 fast trip 2	≈0.80	0.740	0.753
R2 reclose 2	≈1.00	0.940	0.953
Fuse F671-2 melts	–	1.248	0.752
Fuse operation / clears	1.21	1.613	1.253
Fuse heat after the fast shots	–	45 %	105 %

Table C.9 Reclosing sequence of the EMT simulation against the paper's Fig. 10.

DG penetration (Fig. 7 of the paper)

The paper's Fig. 7 gives the CTI at nodes 633 and 671 as shares of a doughnut chart. The trend is the same in both: the CTI falls monotonically as the penetration rises. At 671 the shares of the first four levels are close to the paper's. The levels at which the CTI changes sign differ: in the paper both nodes cross zero at about 40–50 per cent; in this study 633 crosses at about 4 per cent and 671 does not cross up to 100 per cent. The crossing depends on the starting margin, which depends on fuse sizes and settings that the paper does not publish.

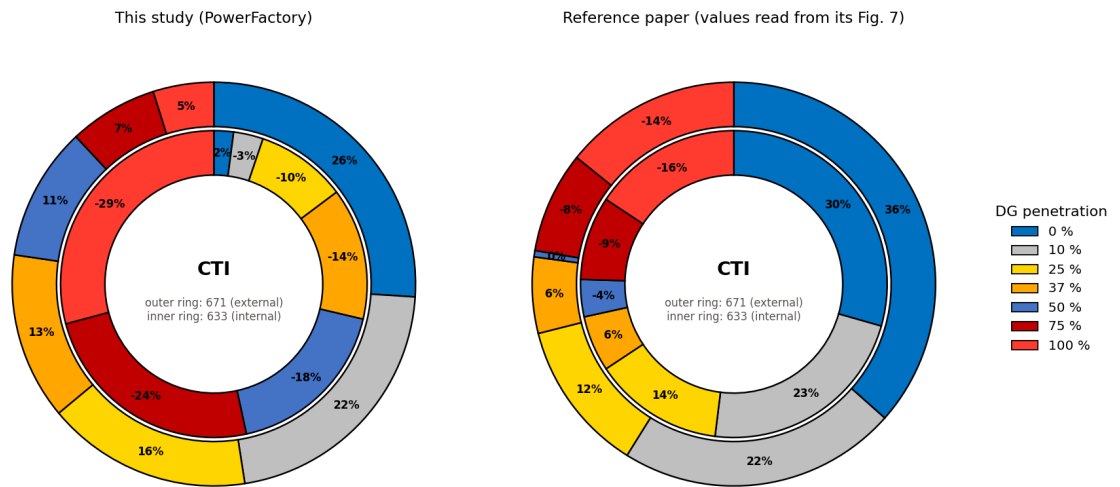


Figure C.1 CTI against DG penetration: this study (left) and the values read from the paper's Fig. 7 (right).

DG (%)	Node 633 (ring %)		Node 671 (ring %)	
	this study	paper	this study	paper
0	+2.1	+30	+26.0	+36
10	-3.1	+23	+21.5	+22
25	-9.6	+14	+16.4	+12
37	-13.9	+6	+13.4	+6
50	-17.9	-4	+10.7	+0
75	-24.2	-9	+7.1	-8
100	-29.2	-16	+4.9	-14

Table C.10 CTI shares against DG penetration: this study and the paper's Fig. 7.